

is doing. We suppose that the agent is a salesman, who will be representing the principal to a particular client. There are three possible outcomes to this interaction: The client can place no order with the principal; the client can place an order that is worth a (gross) \$100 with the principal; or the client can place an order that is worth a (gross) \$400 with the principal. The agent's level of effort affects the odds of each of these three outcomes. If the agent works hard at making the sale, then a \$400 sale results with probability .6, a \$100 sale with probability .3, and no sale with probability .1. If the agent doesn't work hard, there is a \$400 sale with probability .1, a \$100 sale with probability .3, and no sale with probability .6. The size of the sale is observable, and (we assume) the agent's wages can be made contingent upon this variable.

The principal is risk neutral. Note, in particular, where the \$70 and \$270 figures came from; these are the expected gross profits from the sale, for low- and high-effort levels, respectively. Note also that, unless the principal can observe the effort level of the agent directly, the data received (size of the order placed, if any) do not tell conclusively what level of effort was put in. A \$400 sale indicates that a high level of effort was more likely, but it isn't conclusive.

Case 1. A risk neutral agent

Now imagine that the agent is also risk neutral. (Note well: This isn't what we assumed above, and we'll get back to our earlier assumptions in a bit.) By this we mean that the agent's utility function is $u(w, a) = w - a$. For the duration of this case, we assume that the agent's reservation level of utility is 81, and high and low effort correspond to $a = 25$ and $a = 0$, respectively. With these new numbers, the same problem as we had before presents itself: The principal would be willing to hire the agent and pay him a bit more than \$106 if hard work could be guaranteed. This would leave the principal with a net profit of $\$270 - \$106 = \$164$. But the principal would be unwilling to expend the \$81 it would take to get the agent to work if the agent puts in low effort. And she would certainly be unhappy if she hired the agent for \$106 and then he put in low effort.

But now there is a simple solution. Offer the agent the following contingent contract: "If you make no sale, you pay me \$164. If you make a small sale (worth \$100 gross), you only pay me \$64. And if you make a large sale, you will be paid $\$400 - \$164 = \$236$." The agent, offered this, can choose one of the following three courses of action:

(a) Turn down the contract, and get reservation utility level 81.

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(b) Take the contract and put in a low level of effort. This will net expected utility

$$(.1)(236) + (.3)(-64) + (.6)(-164) - 0 = -94.$$

(c) Take the contract and work hard. This will net expected utility

$$(.6)(236) + (.3)(-64) + (.1)(-164) - 25 = 81.$$

The agent is just indifferent between options (a) and (c), and if the principal sweetens the contract just a bit, the agent will prefer (c). The principal is quite happy with this. The agent, in his own interests now, will work hard. Indeed, the principal's net from the sale net of the payment to the agent is \$164 (less any sweetening) with certainty!

What our principal has done is to get the agent to internalize the effect of his effort decision. The agent now bears fully the cost of putting in less than a high level of effort.

Case 2. A risk averse agent

Now go back to the original formulation, where the agent's utility function is $u(w, a) = \sqrt{w} - a$, his reservation utility level is 9, and $a = 5$ for high effort and $a = 0$ for low. If we could write a contract contingent on the effort level of the agent, then the best contract for the principal to write is one in which the agent gets \$196 (plus a bit, perhaps) if he works hard and some low amount (such as \$0) if he doesn't. This leaves the principal with an expected net profit of $\$270 - \$196 = \$74$.

But we assume that the principal can only make the agent's wages contingent on the (gross) size of the sale. In case 1, we could still find a contract that would make the principal as well off as if she could write a contract contingent on actual effort level. But in this case we cannot. Two countervailing forces are at work in this case:

(a) In this case, where the principal is risk neutral and the agent is risk averse, the most "efficient" arrangement is one in which the agent's wage is certain. Why? In general, if one party to a transaction is risk averse and the other is risk neutral, then it is efficient for the risk neutral party to bear all the risks. In the somewhat different context of syndicate theory in chapter 5, you saw this proved formally. The same formal techniques work here. So instead of subjecting you for a second time to the formal proof, let us give the intuition: If the principal pays the agent a random wage, then

the agent evaluates the wage according to his expected utility. Being risk averse, if the wage is at all risky the agent values it at less than its expected value. But the principal, being risk neutral, values the cost of the wages paid at their expected value. If we imagine that the agent's wages had expected value $\bar{w}$, then the principal would see this as an outflow from her pocket equivalent to $\bar{w}$, but the agent would see this as an inflow to his pocket of something less than $\bar{w}$ as long as there is any risk at all.

(b) On the other hand, if we give the agent a riskless wage, the agent has no incentive to work hard. And if the agent doesn't work hard, the principal doesn't want to enter the transaction.

(To induce the agent to work hard, we will have to give up some of the efficiency that is obtained by putting all the risk on the principal. The question is, How can we do this as efficiently as possible?)

To answer this, suppose that we form a contract in which the agent is paid $\$x_0$ if no sale is obtained, $\$x_1$ if the small sale is made, and $\$x_2$ if the large sale is made. I am squaring the values so that when I apply the agent's utility function his utility in each contingency will be $x_i - a$, for $i = 0, 1, 2$. Hence, offered this contract, the agent has three choices:

- (a) Refuse the contract, and get reservation utility 9
- (b) Take the contract and put in a low level of effort, for an expected utility of

$$(.6)x_0 + (.3)x_1 + (.1)x_2$$

- (c) Take the contract and put in a high level of effort, for an expected utility of

$$(.1)x_0 + (.3)x_1 + (.6)x_2 - 5$$

Assume for the moment that we wish to write the best possible contract (from the point of view of the principal) subject to the constraints that the agent will take the contract and put in a high level of effort. Then we wish to

$$\text{minimize } (.1)x_0^2 + (.3)x_1^2 + (.6)x_2^2$$

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subject to

$$(.1)x_0 + (.3)x_1 + (.6)x_2 - 5 \geq 9$$

and

$$(.1)x_0 + (.3)x_1 + (.6)x_2 - 5 \geq (.6)x_0 + (.3)x_1 + (.1)x_2.$$

That is, we wish to minimize the expected wage (since we are taking the perspective of the principal) subject to two constraints; the first is that the agent should sign on the dotted line, and the second is that the agent should then choose to put in the high level of effort. (We should add constraints that the x_i must all be nonnegative, but I'll proceed without them and add them in later if necessary.) These two constraints have names: The first is often called the *individual (rationality or participation) constraint*, and the second is called the *incentive constraint*.

You should have no difficulty solving this constrained optimization problem. To spare you all that needless effort, let us simply give the solution here: $x_0 = 5.42857$, $x_1 = 14$, $x_2 = 15.42857$. Both constraints bind at the optimum, which is fairly intuitive: The principal doesn't want to pay the agent any more than necessary to get him to work, and she doesn't want to put any more risk on the agent than is necessary to get the agent to work hard, because it is costly to her to put risk on the agent. Thus we have the following as the optimal contract, if our objective is to get the agent to take the job and to put in high effort:

If no sale is made, wages are $5.42857^2 = \$29.46$

If a \$100 sale is made, wages are $14^2 = \$196$

If a \$400 sale is made, wages are $15.42857^2 = \$238.0407$

The expected wage bill is $(.1)(29.46) + (.3)(196) + (.6)(238.0407) = \204.56 , which leaves the principal with an expected profit of $270 - 204.56 = \$65.44$.

Compare this with the "first best" contract — the contract where the principal gives the agent a flat wage of \$196 and relies on trust or the compulsion of a monitoring scheme to ensure that the agent puts in a high level of effort. To give the agent the right incentives, we had to have him bear some of the risk by rewarding him in case of the outcome that is more likely if he puts in greater effort. This cost the principal an expected \$8.56.

16.3. Finitely many actions and outcomes

A general formulation

The technique just used generalizes very nicely to a class of principal-agent problems. We imagine an agent who may agree to undertake a task for a principal, and who then chooses an action a to take out of some finite set $A = \{a_1, \dots, a_N\}$. The action choice by the agent is not observed by the principal; instead the principal sees an imperfect signal of what the agent did. We model this by saying that the principal (and the agent) observe a signal s that is drawn from a finite set $S = \{s_1, \dots, s_M\}$. If the agent chooses action a_n , the probability that signal s_m is produced is π_{nm} , where $\sum_{m=1}^M \pi_{nm} = 1$ for each n . The principal is unable to write a contract that makes the agent's compensation directly dependent on a ; the best she (the principal) can do is to make his compensation a function of s .

Note carefully that we refer to the agent's choice of action instead of his choice of effort. We do not preclude the interpretation of a as a level of effort, and in a later subsection we will specialize to a case that has that interpretation quite naturally. But for now we don't rule out other interpretations.

Both for ease of exposition and for some of the results we later give, we make our first assumption:

Assumption 1. The probability $\pi_{nm} > 0$ for all n and m .

In words, every outcome is possible under every action.

The agent's utility depends on the wages he receives, denoted by w , and the action he takes, denoted by a . His preferences over lotteries concerning his income obey the von Neumann-Morgenstern axioms, and $U(w, a) = u(w) - d(a)$ is his von Neumann-Morgenstern utility function. Note carefully that we assume that $U(w, a)$ is additively separable into a piece that depends on wages, $u(w)$, and a piece that depends on the action selected, $-d(a)$. (The letter d here is a mnemonic for disutility.) We assume that the agent has a reservation utility level u_0 . And we add the following innocuous assumption.

Assumption 2. The function u is strictly increasing, continuously differentiable, and concave.

Concavity of u is just risk aversion for our agent (in terms of lotteries over his wages). We don't preclude that u is linear.

The principal cares about the action chosen by the agent and about the wages she pays to him. Specifically, we suppose that $B(a)$ for some function B gives the gross benefits to the principal of hiring the agent if the agent chooses action a , and the principal's net benefit is $B(a)$ less the expected wages she must pay.

This formulation is far from general. We are assuming that the principal is risk neutral, and we assume a very special form of utility function for the agent. Much of what follows can be extended somewhat to encompass more general formulations. In particular, Grossman and Hart (1983), from whom a lot of what follows is taken, assume throughout that the agent's utility function takes the somewhat more general form $U(w, a) = f(a)u(w) - d(a)$ for a strictly positive function f . You will be asked to look at more general formulations in the problems.

Solving the basic problem

The basic problem is to find the optimal incentive scheme for the principal to offer the agent. To solve this problem, we proceed as follows.

Step 1: For each $a_n \in A$, what is the cheapest way to induce the agent to take the job and choose action a_n ? Cheapest here is measured in terms of the expected wages that must be paid. Following the pattern of our example from the previous section, we solve this problem by solving a constrained maximization problem.

The variables in this maximization problem are the levels of "wage-utility" the agent is given as a function of the signal s . That is, we take variables x_m for $m = 1, \dots, M$, where if $w(s_m)$ is the wage paid to the agent if the signal is s_m , then

$$x_m = u(w(s_m)).$$

We assume the $u(\cdot)$ is a strictly increasing and continuous function, and we let v be the inverse of u ; that is, $v(z) = w$ if $u(w) = z$. Thus the wage paid to the agent if signal s_m is produced, as a function of the variable x_m , is just

$$w(s_m) = v(x_m).$$

Thus, as a function of the variables $\{x_1, \dots, x_M\}$, the expected wages the principal must pay if the agent takes action a_n is

$$\sum_{m=1}^M \pi_{nm} v(x_m).$$

If the agent is offered wages as a function of signal as given by $v(x_m)$, what constraints must be met to be sure that he will select action a_n ? We first must be sure that in choosing a_n the agent achieves at least his reservation level of utility,

$$\sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq u_0.$$

Note two things here. First, the expected utility of wages to the agent is $\sum_{m=1}^M \pi_{nm} u(v(x_m))$, which since v is the inverse of u is just $\sum_{m=1}^M \pi_{nm} x_m$. Second, we have a weak inequality, which means that if the constraint is binding the agent is indifferent between taking the job or not. It is standard to work with weak inequalities, presuming that the agent, if indifferent, will resolve ties in the interests of the principal. (See the later subsection on game theoretic connections.)

We must be sure that choosing a_n is better than choosing some other action n' . This is modeled by imposing the constraints

$$\sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq \sum_{m=1}^M \pi_{n'm} x_m - d(a_{n'}), \quad n' = 1, \dots, N.$$

The two comments made in the previous paragraph apply here as well. Note that we have included the constraint for $n' = n$, although it is satisfied trivially.

There may also be constraints on the level of wages that can be paid. A standard constraint of this sort is that wages may be constrained to be nonnegative. For example, this was implicit in our example, since the agent's square-root utility function is not defined for negative wages. In such cases we would add constraints such as $x_m \geq v(0)$. We will not carry constraints such as this along in our formulation, although at one point we will comment on the effect that such a constraint might have.

So we have step 1: For each action a_n

$$\begin{aligned} & \text{minimize } \sum_{m=1}^M \pi_{nm} v(x_m) \\ & \text{subject to } \sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq u_0, \text{ and} \\ & \sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq \sum_{m=1}^M \pi_{n'm} x_m - d(a_{n'}), \quad n' = 1, \dots, N. \end{aligned}$$

Call the value of this problem (that is, the value of the objective function at the optimal solution) $C(a_n)$. This is the *minimal expected cost of inducing the agent to select action a_n* . We will use the label $C(n)$ for this problem, a mnemonic for cost of action number n . The first constraint is called the *participation* constraint, and the other constraints are called the *relative incentive* constraints.

For readers who know about such things: Because u is concave, v is convex, and this is a well-behaved mathematical programming problem. We are minimizing a convex function subject to linear constraints, so that satisfaction of the first-order equations and the complementary slackness conditions (and the problem constraints) is necessary and sufficient for a solution. General convex programming algorithms can be employed to solve this problem numerically.

For a given a_n , there may be no solution at all on grounds that the set of values (x_m) that meet all the constraints is empty; examples are easy to construct. In this case we would say that $C(a_n) = +\infty$. Note that $C(a_n)$ is finite for at least one n . If we set $x_m \equiv u_0 + \min_n d(a_n)$, then the constraints are all satisfied for the problem (Cn^*) where n^* is the index of the effort level that has minimal disutility. In fact, for cases where u is concave this is the solution to (Cn^*) , a result that you should find easy to prove following chapter 5 or from the first-order equations. (See later.)

The problem of nonexistence of any solution to the constraints is the only sort of problem concerning existence of a solution to (Cn) that may arise: If there is some set of variables (x_m) , which meets all the constraints, then there is an optimal solution. You are asked to show this in problem 9; hints as to how to prove this are given there. It is worth noting that this result depends crucially on the assumption that $\pi_{nm} > 0$ for all n and m ; see problem 7.

Step 2. For which $a \in A$ is $B(a) - C(a)$ maximized? This is a simple maximization problem.

Since we know that $C(a_{n^*})$ is finite where n^* is as before, and since each $C(a_n)$ is either finite or equal to $+\infty$, there is always a solution to the principal's overall problem.

If we wish to be very careful about this, we have to wonder what happens if $B(a) < C(a)$ for all a . Is it viable for the principal to refuse to hire the agent at all or, rather, for the principal to make a wage offer that the agent is sure to turn down? If so, what are the consequences for the gross benefits of the principal? We will not be tidy about this possibility but instead implicitly assume that some level of effort a can be implemented at a cost sufficiently low to make it worth the principal's while to do so.

The key to this technique is the way that it takes the problem in steps. First we find the minimum cost way to induce action a for each $a \in A$, and then we choose the optimal a by comparing benefits and costs.

Basic results and analysis

Within the context of this somewhat general formulation, we can obtain a result along the lines sketched in the previous section.

To set a benchmark, consider the solution to the problem if it is possible to specify (and enforce the choice of) the action in the contract. From results in chapter 5, since the agent is risk averse (possibly risk neutral) and the principal is risk neutral, the cheapest method of guaranteeing the agent a given level of utility is with a fixed payment. (If the agent is risk neutral over some range, other schemes may be equally cheap, but none will be cheaper.) So the cheapest way to get the agent to accept a contract that specifies action choice a_n is to offer to pay him

$$C^0(a_n) \equiv v(u_0 + d(a_n)).$$

To be very formal about this, this is obtained from the solution of the problem

$$\text{minimize } \sum_{m=1}^M \pi_{nm} v(x_m) \text{ subject to } \sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq u_0.$$

This is the "same" as the problem (Cn) except that the relative incentive constraints are missing. We don't need them because the contract, by assumption, can specify which action is taken. Since the problem is just like (Cn) but with some constraints eliminated, it is evident that $C^0(a_n) \leq C(a_n)$ for all n . We can go on to

$$\text{maximize } B(a_n) - C^0(a_n);$$

the effort level that solves this problem is called the *first-best level of effort*.

We are supposing, however, that the principal can't write an enforceable contract that specifies the first-best or any other level of effort. As in the example of the previous section, we obtain the following result:

Proposition 1. *If the agent is strictly risk averse, $C^0(a_n) < C(a_n)$ for any action a_n that is more onerous than some other action; i.e., such that $d(a_n) > \min_{a'} d(a')$.*

The proof is fairly simple. If the agent is strictly risk averse, then the unique efficient risk-sharing arrangement is for the principal to bear all

the risk; the agent gets a sure wage. But if the agent is given a wage that is independent of the signal, he will choose the least onerous action.⁴

We next address the question *Is the participation constraint binding at the optimal solution?* The answer is (a qualified) yes. In fact, for any effort level a_n such that the constraints defining (Cn) can be satisfied, in the solution of (Cn) the participation constraint is binding (with qualifications). To see why, suppose that (x_m) is a solution of (Cn) such that $\sum_{m=1}^M \pi_{nm} x_m - d(a_n) = u' > u_0$. Consider the alternative incentive scheme given by $x'_m = x_m - u' + u_0$. This clearly satisfies the participation constraint (with equality!), and since we decrease each x_m by the same amount, satisfaction of the relative incentive constraints is guaranteed. But if v is strictly increasing, this lowers the cost to the principal, which contradicts the assumption that (x_m) is a solution of (Cn). Nothing very mysterious is going on here. We simply are taking the same amount of wage-utility away from the agent at every outcome, which doesn't affect his relative incentives at all.

What is the qualification? Suppose there were constraints on the variables x_m such as $x_m \geq u(0)$. Then we could not be sure that x'_m , which is x_m less some amount, would continue to satisfy such constraints. In fact, it is easy to construct examples where, because of the presence of constraints such as $x_m \geq u(0)$, the participation constraint does not bind. To be sure that this problem doesn't appear, we need to be sure that we are always able to lower the utility levels received by the agent; essentially, u should have range that is unbounded below. (If the domain of u is all of R , this will hold automatically since u is concave. So this qualification pertains to cases where the domain of u is itself bounded below.)

As a final piece of basic analysis, we turn to the first-order conditions (and complementary slackness conditions) of the problem (Cn). Our reason for interest in these conditions is straightforward. As we noted before (in the small type), assumption 2 implies that satisfaction of the first-order and complementary slackness conditions of (Cn) is necessary and sufficient for an optimal solution of (Cn).

Letting λ be the multiplier on the participation constraint and η_n be the multiplier on the relative incentive constraint for action a_n , the

⁴ In the previous section we had something of a converse to this proposition, namely that if the agent is risk neutral, then $\max_a B(a) - C^0(a) = \max_a B(a) - C(a)$; the principal can achieve her first-best outcome. We don't obtain that result in this general formulation; it must await a specialization to be given in the next subsection.

first-order condition for x_m is

$$v'(x_m) = \lambda + \sum_{n'=1}^N \eta_{n'} \left(1 - \frac{\pi_{n'm}}{\pi_{nm}} \right).$$

(This is after some algebraic manipulation, which you should replicate.) This has a very clean and intuitive explanation. Recall that the multipliers must all be nonnegative and, from the complementary slackness conditions, $\eta_{n'}$ will be strictly positive only if the relative incentive constraint on action $a_{n'}$ is binding. Note that since v is convex, v' is an increasing function. Thus a larger right-hand side in the first-order condition means a larger value of x_m . With these preliminary observations, think of the scheme (x_m) as follows:

(a) A “base payment” (measured in utility) is made to the agent, given by the equation $v'(x_m) = \lambda$. If none of the relative incentive constraints bind, then this is the first-order equation, and we have the result that x_m is constant in the outcome m .^a This is not surprising; we are just rediscovering the result from chapter 5 that when there are no incentive problems the optimal arrangement gives the risk-averse agent a constant wage.

(b) But when there are binding relative incentive constraints, the wages of the agent are not constant. Specifically, if $\eta_{n'} > 0$, then the right-hand side, and hence the agent's wages, are increased at outcomes m such that $\pi_{n'm}/\pi_{nm} < 1$ and they are decreased at outcomes m such that $\pi_{n'm}/\pi_{nm} > 1$. When $\pi_{n'm}/\pi_{nm} < 1$, outcome m is less likely if the agent takes action n' than if he takes the desired action n . So this does indeed seem a good time to pay him more to get him to choose n over n' . And the agent is “penalized” if the outcome is an s_m such that $\pi_{n'm}/\pi_{nm} > 1$, an outcome that is more likely under the action choice n' than under the desired n .

That, pretty much, is all there is to it. Of course, the levels of the “basic marginal utility wage” λ and the relative effort multipliers $\eta_{n'}$ all must be determined, and they must work together in a fashion that causes all the initial constraints and the complementary slackness conditions to be met. But the essential idea is simple: At the optimum, you reward the agent if the outcome is relatively more likely if he took the desired action,

^a To be precise, it is constant if u is strictly concave, hence v is strictly convex, and hence v' is strictly increasing.

and you penalize him if the outcome is relatively less likely, relative to the actions that bind in the relative incentive constraints.

The formulation: A special case

We next make use of a special case of the formulation given above. To begin, we imagine that the signals s_m are levels of gross profit accruing to the principal, exactly as in the example of the previous section. In this case $B(a_n) = \sum_{m=1}^M \pi_{nm} s_m$.

With this specialization, we can give the converse to the proposition from the previous subsection:

Proposition 2. For a risk neutral agent, $\max_a B(a) - C^0(a) = \max_a B(a) - C(a)$. In fact, if a^* achieves the maximum in $\max_a B(a) - C^0(a)$, one scheme that implements this action is for the principal to pay the agent $s_m - B(a^*) + C^0(a^*)$ if the gross profits are s_m , so the principal receives $B(a^*) - C^0(a^*)$ for sure and the agent bears all the risk.

We will not go through the proof of this result except to note the intuition: If the agent is risk neutral, efficient risk sharing between the principal and agent is consistent with the agent bearing all the risk. And by having the agent bear all the risk, we have him bearing entirely the consequences of his action choice. It is as if the principal “sold the venture” to the agent, who is now sole proprietor and is working for himself, and who now chooses the optimal action in his own sole and best interests.^b

To continue our specialization, we further imagine that the actions are levels of effort that the agent might select, ordered in terms of increasing disutility. That is,

$$d(a_1) < d(a_2) < \dots < d(a_N).$$

In this case, there is no reason to carry around the extra baggage of the function d ; we can use a_n for the disutility of effort choice n , so that $U(w, a) = u(w) - a$, and the condition becomes $a_1 < a_2 < \dots < a_N$.

Finally, we imagine that higher (i.e., more onerous) levels of effort result in higher gross profits for the principal. To write this assumption, we assume that the gross profit levels s_m have been arranged in increasing

^b Perhaps now is the time to ask, Why couldn't we give this proposition before we specialized to this case? If the answer isn't obvious, consider what would happen in the previous subsection if the number of signals, M , was one.

order: $s_1 < s_2 < \dots < s_M$.^c To assume that higher levels of effort lead to higher profits, we might simply assume that $\sum_{m=1}^M \pi_{nm} s_m$ is increasing in n . But usually something more stringent is assumed, namely that an increase in effort results in a higher probability of higher levels of profit. Formally:

Assumption 3. For all pairs n and n' such that $n' > n$ and for all $m = 1, \dots, M$,

$$\sum_{i=m}^M \pi_{ni} \leq \sum_{i=m}^M \pi_{n'i}$$

with for each n and n' a strict inequality for at least one m .

For those who know the terminology, assumption 3 states that increased effort results in a first-order stochastic increase in the levels of gross profits. Although it may not be immediately evident to you, this does imply that $\sum_{m=1}^M \pi_{nm} s_m$ is increasing in n . (See problem 10.)

We motivate the analysis to come by asking the following question:

Is the wage paid, or equivalently the wage-utility level x_m , nondecreasing with increases in the level of gross profits?

Since the principal will wish to reward the agent for higher levels of effort, and since higher levels of effort “go with” higher levels of profit, it may seem natural to suppose that the answer will be yes. But the answer is no in general. An easy-to-comprehend example where it fails runs as follows.

There are three possible levels of gross profit, \$1, \$2, and \$10,000, and two possible effort levels, $a_1 = 1$ and $a_2 = 2$. (Remember that we identify an effort level with its disutility in this specialized formulation.) If the agent chooses effort level a_1 , the outcomes are \$1 gross profit with probability .5, \$2 gross profit with probability .3, and \$10,000 gross profit with probability .2. If the agent chooses effort level a_2 , the outcomes have probabilities .4, .1 and .5, respectively. Note that as promised, higher effort leads to higher probabilities of better outcomes in precisely the sense of assumption 3.

We leave it to you to flesh out this example so that the agent has a strictly concave utility function for wages u and a reservation utility level u_0 such that the solution to the principal's problem is to provide the agent

^c The use of strict inequalities here may trouble the very picky reader. In essence, we are assuming that the *only* information upon which contracts can be written is the level of gross profits.

with incentives to choose a_2 . This shouldn't be hard to do. Expected gross profits if the agent chooses a_1 are \$2,001.50, while they are \$5,000.60 if he chooses a_2 . This gives the principal a strong reason to create incentives for a_2 .

When you do this, the multiplier on the relative incentive constraint for choosing a_2 instead of a_1 will necessarily be strictly positive. Why? Because if it isn't, in the first-order conditions we will get a flat wage payment to the agent, and we know that this will cause him to choose a_1 . (Go slowly here. I'm skipping steps to force you to think this through.) So the three first-order equations will be

$$v'(x_1) = \lambda + \eta(1 - (.5/.4)) = \lambda - .25\eta$$

$$v'(x_2) = \lambda + \eta(1 - (.3/.1)) = \lambda - 2\eta, \text{ and}$$

$$v'(x_3) = \lambda + \eta(1 - (.2/.5)) = \lambda + .6\eta,$$

where x_1 is the wage-utility level corresponding to \$1 gross profits, x_2 corresponds to \$2, and x_3 corresponds to \$10,000, λ is the multiplier on the participation constraint, and η is the multiplier on the binding relative incentive constraint. The point, simply, is that if $\eta > 0$, which it must be, then $x_1 > x_2$.

There should be no mystery why this is. While higher effort shifts upwards the probability of higher gross profits in the sense of assumption 3, the ratio of the probability of the *outcomes* under a_2 to their probability under a_1 goes down and then up. Evidently, to get an affirmative answer to the question posed above, we need something stronger than assumption 3.^d We seemingly need

Assumption 4. The monotone-likelihood ratio property. For any two effort levels a_n and $a_{n'}$ such that $n < n'$, and for any two gross-profit outcomes s_m and $s_{m'}$ such that $m < m'$, the relative likelihood of the better outcome under the higher effort level to the lower is at least as large as this likelihood ratio for the lower outcome. Or, in symbols,

$$\frac{\pi_{n'm'}}{\pi_{nm'}} \geq \frac{\pi_{n'm}}{\pi_{nm}}$$

^d You should verify that assumption 4, which follows immediately, is indeed strictly stronger than assumption 3. In fact, if you know about first-order stochastic dominance, you might wish to ponder the connections between that property and assumption 4.

Is assumption 4 enough to get an affirmative answer to our motivating question? It is certainly moving us in the right direction, but even this it is not enough. To see why, go back to the first-order equations for (Cn), for the effort level a_n that the principal optimally chooses to implement. Insofar as the only binding relative incentive constraints are for effort levels less than a_n , we are okay. Because v is convex, the sign of $x_m - x_{m'}$ for $m < m'$ is the same as the sign of $v'(x_m) - v'(x_{m'})$. But from the first-order conditions,

$$v'(x_m) - v'(x_{m'}) = \sum_{n'=1}^N \eta_{n'} \left[\frac{\pi_{n'm'}}{\pi_{nm'}} - \frac{\pi_{n'm}}{\pi_{nm}} \right].$$

As long as $n' < n$, assumption 4 says that the sign of each term in the brackets is nonpositive, hence (since multipliers are all nonnegative) the sum is nonpositive, which is just what we want.

But if $\eta_{n'}$ is positive for some $n' > n$, then we may be sunk; now the monotone likelihood ratio property runs the wrong way.

It may seem incredible to you that, in any problem (Cn), the relative incentive constraints for greater levels of effort than the one desired will be binding. Or rather, it may seem incredible that this would hold in the problem (Cn) for the a_n that is optimal for the principal. After all, if the agent wants to choose a higher level of effort, why would the principal ever wish to stop him from doing so? There is a reason why. It is true that the principal benefits from higher levels of effort by the agent in terms of the function $B(a)$. But for a given wage schedule, there are also costs that the principal incurs: Insofar as higher wages are paid for better outcomes, by expending more effort the agent increases his expected wages. It is possible to construct examples where there are three levels of effort, a_1 , a_2 , and a_3 , the monotone likelihood ratio property holds, the middle level of effort is optimal, and, in the problem (C2) both relative incentive constraints bind in a way that gives optimal wages that are not nondecreasing in gross profits.^e The details of this example are not illuminating; if you wish to see them, consult Grossman and Hart (1983, example 1).

What is needed in addition to assumption 4 is an assumption that implies that the only binding relative incentive constraints in (Cn) for the optimal effort level a_n are constraints corresponding to levels of effort lower than a_n . We proceed to develop one condition, which Grossman

^e Test your understanding of this by answering the following question before reading further. Why is it impossible, in this case, that only the relative incentive constraint for a_3 binds in (C2)?

and Hart attribute to unpublished work by J. Mirrlees. (This condition will play a role in later analysis as well, which is the excuse for not shifting to the smaller type size just yet.)

Assumption 5. If a_n , $a_{n'}$, and $a_{n''}$ are effort levels such that $a_{n'} = \beta a_n + (1 - \beta)a_{n''}$, then for each $m = 1, \dots, M$,

$$\sum_{i=m}^M \pi_{n'i} \geq \beta \sum_{i=m}^M \pi_{ni} + (1 - \beta) \sum_{i=m}^M \pi_{n''i}.$$

This is called the *concavity of the distribution function condition*. To aid in making comparisons with the literature and, in particular, with the treatment in Grossman and Hart (1983), recall that here we are identifying the effort level a_n with its disutility. To interpret this assumption (and for later purposes of notation), let $\Pi_m(a_n) = \sum_{i=m}^M \pi_{ni}$. That is, $\Pi_m(a_n)$ is the probability of seeing a gross profit level at least as large as s_m if effort level a_n is taken. Then, roughly, the assumption says that increases in effort (as measured by their disutility) have decreasing marginal impact on the probabilities of better outcomes. You can see this most easily if you think of a_n , $a_{n'}$, and $a_{n''}$ such that $a_n - a_{n'} = a_{n'} - a_{n''} > 0$. This gives $\beta = 1/2$ in the assumption. Then the increase in disutility in moving from the lowest level of effort $a_{n''}$ to the intermediate level $a_{n'}$ is the same as the increase in moving from $a_{n'}$ to a_n . And according to the assumption, for each m , $\Pi_m(a_{n'}) - \Pi_m(a_{n''}) \geq \Pi_m(a_n) - \Pi_m(a_{n'})$.

What does this assumption do for us?

Proposition 3. If assumptions 1, 2, 4, and 5 all hold and u is strictly concave, then the optimal wage-incentive scheme for the principal has wages that are nondecreasing functions of the level of gross profits.

The proof of this proposition is a bit much for one's first pass through this subject, so we relegate most of the details to smaller type. But before switching type size, we record the first step along the way, which we will need in the next section. We first give a piece of notation. For any fixed wage-incentive scheme (w_m) , let $U(a_n)$ be the overall expected utility of the agent under this scheme if he chooses effort level a_n . That is,

$$U(a_n) = \sum_{m=1}^M \pi_{nm} u(w_m) - a_n.$$

Lemma. Fix a wage-incentive system that is nondecreasing in the gross profit of the firm. (That is, $w_1 \leq w_2 \leq \dots \leq w_M$.) If assumption 5 holds, $U(a)$ is a concave function of a .⁵

Note carefully: This is true of any nondecreasing wage-incentive system. As you will see in the proof, it doesn't depend at all on an assumption 4 or even on an assumption 3. And all that is required from assumption 2 is that u is nondecreasing.

Proof of the lemma. Write x_m for $u(w_m)$. If the function u is nondecreasing (which is assumed), then so is the sequence x_m . Write $\delta_1 = x_1$ and, for $m = 2, \dots, M$, let $\delta_m = x_m - x_{m-1}$. Note that $\delta_m \geq 0$ for all $m \geq 2$. Then

$$\begin{aligned} \sum_{m=1}^M \pi_{nm} u(w_m) &= \sum_{m=1}^M \pi_{nm} x_m = \sum_{m=1}^M \left[\pi_{nm} \sum_{j=1}^m \delta_j \right] \\ &= \sum_{m=1}^M \left[\delta_m \sum_{i=m}^M \pi_{ni} \right] = \delta_1 + \sum_{m=2}^M \delta_m \Pi_m(a_n). \end{aligned}$$

The first, second, and fourth equalities are just a matter of rewriting; it is the third equality that may not be immediately obvious. But if you consider the matter carefully, you will see that this is correct. Hence

$$U(a_n) = \delta_1 + \sum_{m=2}^M \delta_m \Pi_m(a_n) - a_n.$$

This is the sum of a constant, nonnegative weightings of concave functions of a_n , and a linear function of a_n . Hence it is concave in a_n .

With this lemma in hand, we can proceed to a second lemma. To set up for this, consider, for each $n = 1, 2, \dots, N$, the problem

$$\begin{aligned} &\text{minimize } \sum_{m=1}^M \pi_{nm} v(x_m) \\ &\text{subject to } \sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq u_0, \text{ and} \\ &\quad \sum_{m=1}^M \pi_{nm} x_m - d(a_n) \geq \sum_{m=1}^M \pi_{n'm} x_m - d(a_{n'}), \quad n' = 1, \dots, n-1. \end{aligned}$$

⁵ The notion of a concave function on a discrete domain may be a bit mysterious to you, but the idea is that if $a' = \beta a + (1 - \beta)a''$ for a, a' , and a'' all in the domain of U and $\beta \in [0, 1]$, then $\beta U(a) + (1 - \beta)U(a'') \leq U(a')$.

To see the difference between this and (C_n) , look closely at the range of n' in the third line. This problem is just the same as (C_n) , except that the relative incentive constraints for effort levels greater than n are missing. We will call this problem (C^*n) .

Lemma. If assumptions 1, 2, and 4 hold and u is strictly concave, then any solution (x_m) of (C^*n) is nondecreasing: $x_1 \leq x_2 \leq \dots \leq x_M$. If, in addition, assumption 5 holds, then any solution of (C^*n) is a solution of (C_n) .

Proof. A necessary condition for a solution of (C^*n) is the first-order conditions. If u is strictly concave, the values of the x_m at any solution are strictly increasing in the right-hand sides of the first-order equations. And in (C^*n) , as long as assumption 4 holds, those right-hand sides are nondecreasing in m . The last assertion follows because, by construction, in (C^*n) we only have relative incentive constraints for effort levels less than the one being implemented. This gives the first assertion of the lemma.

For the second assertion, we consider cases. For $n = 1$, strict concavity of u ensures that the solution of both (C^*n) and (C_n) is a constant wage set equal to reservation wage of the agent. For $n \geq 2$, we know that the solution to (C^*n) involves a nonconstant wage, because at a constant wage a_1 would be chosen. From the first-order conditions this implies that at least one of the relative incentive constraints must have a positive multiplier, which by complementary slackness implies that this constraint must be binding. Let n' be the binding constraint, which of course has $n' < n$. Fix the wage-incentive scheme that solves (C^*n) , and consider, as in the previous lemma, $U(a)$ as a function of a for $j = 1, \dots, N$, at this wage-incentive scheme. By the first assertion of this lemma combined with the previous lemma, $U(\cdot)$ is concave in a . By construction, $U(a_n) = U(a_{n'})$ (and these are at least as large as $U(a_j)$ for every $j \leq n$). But then the concavity of $U(a)$ in a for all a implies that $U(a_j) \leq U(a_n)$ for all $j > n$. To see this, suppose that $U(a_j) > U(a_n)$ for some $j > n$. Since a_n is a strict convex combination of a_j and $a_{n'}$, we would know that $U(a_n)$ must be at least the corresponding strict convex combination of $U(a_j)$ and $U(a_{n'})$, which would contradict $U(a_{n'}) = U(a_n) < U(a_j)$.

Thus we know that at the optimal solution of (C^*n) , $U(a_j) \leq U(a_n)$ for all $j > n$. But this says that the solution of (C^*n) solves the additional constraints imposed in (C_n) , and so it is a solution of (C_n) . This completes the proof.

Proof of Proposition 3. Suppose that the optimal solution of the principal's problem corresponds to implementation of effort choice a_n . Then the solution of (C^*n) , which is a solution of (C_n) , is nondecreasing. Strict concavity of u implies that the solution of (C_n) is unique. (If there are two solutions, take a convex combination of them. All the constraints continue to be met and since v is strictly convex, the expected wage paid at the convex combination is less.) That gives the result.

One can go on to examine conditions under which the optimal wage-incentive schedule is convex or concave as a function of gross profits. Grossman and Hart (1983, proposition 9) give a result along these lines.

A further specialization: The case of two outcomes

Instead of pursuing this, we turn to a different sort of question. So far we have been concerned primarily with properties of the optimal incentive wages. The results we have derived have concerned solution of the subproblem (C_n) without any real use or mention of the larger problem of choosing an effort level to implement. You might wonder what can be said about the level of effort that the principal optimally chooses. An intuitive case might be made that the principal would always choose to implement a level of effort less than or equal to the first-best level, since risk-sharing between the agent and principal entails shielding the (risk-averse) agent from some of the consequences of his effort in terms of profits, while he bears fully its disutility. (If you are sceptical about this "intuition," don't worry; scepticism is appropriate.)

We will examine this issue in a very special setting. As in the previous subsection, we hold to the story of the action choice being effort measured along a one-dimensional scale of disutility, and we continue to assume that the only signal received about what effort the agent chose is the gross profit received. But we further specialize by assuming that gross profits can take on only two possible levels: s_1 and $s_2 > s_1$. In words, we imagine that the agent's efforts result in either *failure* or *success* of some venture, and profit depends solely on whether the venture succeeds.

In this setting, our assumptions translate as follows: Assumption 1 simply says that both success and failure are possible outcomes at any level of effort; assumption 2 is not changed or reinterpreted; assumption 3 is that higher levels of effort lead to increased chances of success, and this implies, with no further restrictions, that assumption 4 holds. That is, the monotone likelihood ratio property is redundant in this special situation, given assumption 3. Finally, assumption 5 states simply that the probability of a success is a concave function of the effort level.

Given assumption 3, we do not require assumption 5 in this case to obtain the result that the optimal wage incentive scheme is nondecreasing in effort if u is strictly concave. To see why, note that with two outcomes the only way one can fail to have a nondecreasing wage-incentive schedule is if failure is rewarded more than success. But given assumption 3, for any such wage-incentive schedule the agent will choose the lowest level of effort. And we know that (for strictly concave u) a better way to implement the lowest level of effort is to have constant wages.

Hence, with assumption 3 alone, we can think of wages in this case as being of a particularly simple form. The agent has a base wage, denoted by b_1 , to which is added a bonus $b_2 \geq 0$ in the event of a successful outcome.

If we write $\sigma = b_2/(s_2 - s_1)$ the agent's wages are b_1 and $b_1 + \sigma(s_2 - s_1)$, or a base wage plus a "share" of any extra gross profits accruing from a success. We know that $\sigma \geq 0$, and we know that $\sigma > 0$ if the principal wishes to implement any effort level above the minimal level possible (why?). We can also bound σ from above.

Proposition 4. *If u is strictly concave, then $\sigma < 1$ at the optimal wage-incentive schedule.*

The idea here is that if $\sigma \geq 1$, then the principal prefers that the outcome is failure, ex post. This can never be part of an overall optimal scheme. The proof is surprisingly difficult and is omitted here. See Grossman and Hart (1983, proposition 4), who gave the result generalized suitably to cases with more than two outcomes.

Now we return to the question with which we began. Can one say anything in general about how the effort level the principal chooses to implement compares with the first-best effort level? An example will illustrate the difficulties.

Example, part 1: There are two outcomes and two possible effort levels. Effort level $a_1 = 0$ gives a .9 probability of failure and .1 of success, while effort level $a_2 = .1$ gives probabilities .85 and .15, respectively. The agent has utility function $U(w, a) = \ln(w) - a$. The reservation utility level is 0. Success is worth \$10 to the principal, and failure is worth \$0.

It is immediate that to implement effort level a_1 , the principal should pay a constant wage of \$1, which nets her \$0; the \$1 gross expected profit just goes to pay the agent's wages.

If the principal could write a contract that specified (in enforceable fashion) effort level a_2 , the optimal contract would give the agent wages of $e^{0.1} = \$1.105$ regardless of action, for an expected net profit of $\$1.50 - \$1.105 = \$0.395$. Hence the first-best effort level is a_2 .

But to implement the first-best level of effort, the principal must pay the agent higher wages in case of success than in case of failure. It is easy to work out the optimal wage-incentive contract. In the problem (C2), both the participation and the sole relative-incentive constraints must bind. (Why?) These are two linear equations in two unknowns and solving them gives wages of \$.8187 if the outcome is failure and \$.604965 in case of success. (You should do the arithmetic if you are at all unsure where these numbers come from.) This has an expected cost (in terms of wages) of \$.16034, which means that the principal's net profit from implementing

a_2 is a loss of \$.1034. Hence the principal chooses to implement a_1 with a flat \$1 wage.

Example, part 2: Now we add a third possible effort level, $a_3 = 2.27$, which gives a .99 chance of success. Note that assumption 5 is satisfied in this case.

First we look at what the principal could make if she could write an enforceable contract that specified a_3 . The optimal wages in such a contract would pay $e^{2.27} = \$9.6794$ regardless of outcome, for a net to the principal of $\$9.90 - \$9.6794 = \$2206$. Hence the first-best effort level remains a_2 .

(Although it isn't germane, we can ask whether the addition of this third action changes the cost of implementing a_2 . If a_2 can be implemented, it must be at wages where the participation and relative-incentive constraint for a_1 bind. (Why?) So we know that the wages must be as in part 1 of this example, and the only thing to check is whether, at these wages, a_2 is better than a_3 for the agent. (What would happen if a_3 turned out better than a_2 at these wages? What would $C(a_2)$ be in that case?) In fact, this does work out — at the wages computed in step 1, actions a_1 and a_2 give the agent expected utility of 0, while a_3 gives him an expected utility of $-.49$.)

What does it cost to implement a_3 ? When solving the subproblem (C3), we know that the participation constraint must bind. We also have two relative-incentive constraints and only two unknowns. So, unless there is a coincidence, one of those two relative-incentive constraints will bind and the other will be slack. (Why can't they both be slack at the solution of [C3]?) If you try each possibility in turn, you will find that it is the relative-incentive constraint for a_2 that binds, giving wages of \$.7501 if the outcome is failure and \$.9327 if the outcome is success. This costs the principal an expected \$.8409 in wages, for a net profit of \$.0591. Hence this is the solution when there are three possible effort levels.

This covers all the possibilities. In the case of only two effort levels, the principal chooses an effort level lower than the first-best. With the third added, she chooses to implement an effort level that is higher than the first-best. (You should have little problem creating an example where she chooses the first-best level itself.) All that remains is to see why this is happening.

The basic idea is that in our example effort level a_2 , while first-best, gives very weak statistical evidence against the hypothesis that the effort

level chosen was a_1 . Relative to what one gains from implementing a_2 in terms of gross profits, the cost of separating a_2 from a_1 is substantial. Effort level a_3 , on the other hand, gives a very different statistical pattern of outcomes, and so it is not so costly to separate from the others, relative to what is gained in gross profits.

To drive this point home, imagine that a_3 ensured success. (This violates assumption 1 and so is only for purposes of discussion.) Then we could get incentive schemes that implement a_3 with costs arbitrarily close to the first-best cost of a_3 . We would do this with dire threats in case the outcome is a failure — wages very close to zero. The agent can avoid the threat by choosing a_3 , if a_3 ensures success. But threats of this sort are terrible in case the agent contemplates any action that has a nonzero chance of failure, since $\ln(w) \rightarrow -\infty$ as $w \rightarrow 0$. For more on this point, see problem 7.

Connecting to game theory

The analysis that we have conducted has not been couched in game theoretic terms, but it is easy enough to do so. The "game" that is being played can be thought of as follows: The principal offers a contract to an agent, who then either accepts or rejects the contract and, if he accepts it, chooses an action to take. The space of contracts that the principal may offer is limited exactly as we have postulated; the payoff of the agent if he rejects the contract is his reservation level of utility; and the payoffs to both principal and agent are computed in obvious fashion.

The solution concept we have been using, at least for the game just described, is subgame perfect Nash equilibrium. Note where we use the subgame perfection: We assume that the agent accepts the contract and chooses his action optimally given the contract. We could easily construct subgame imperfect equilibria where the agent threatens to reject any contract that doesn't give him some amount more than his reservation level of utility. (Or he could threaten to accept such a contract but then take an action that, while suboptimal for him, is disastrous for the principal.) We have implicitly ruled out such threats.

It sometimes troubles readers that in our equilibria the agent is indifferent between many courses of action, and we assume he chooses the one we or, rather, the principal desires. But this is standard fare for this form of analysis. Thinking of this as a game, it is the continuous action space for the principal that forces this in our solutions. If, for example, we tried to require that the principal offer a contract that makes acceptance strictly better than rejection, there would be no optimal contract at all: At

any contract where the participation constraint doesn't bind, we can design a better contract where it still doesn't bind. And there would be no subgame perfect equilibrium of our game. If the agent used the strategy of rejecting contracts that gave him precisely his reservation utility, the principal's better response (assuming she gets positive net profits at the optimal scheme) would be to sweeten the contract just a bit, so the agent isn't indifferent. But there is no optimal amount of sweetener to put in; there is no single best response. The only way we get equilibrium is to assume that ties are broken in a fashion that favors the first mover.^f This is a "problem" with the style of analysis we use that you should learn to live with, because it is pervasive in the literature.

16.4. Continuous actions: The first-order approach

If you consult the early literature on the principal-agent model, you will find models that assume that both the possible signals and the possible actions or effort levels are not finite in number. The typical model will say that the agent can choose any effort level drawn from some interval $[a_0, a_1]$ and that the range of possible gross profit levels is R or some appropriate subinterval.

Some technical difficulties arise if there are infinitely many possible signals or outcomes. These are especially associated with assumption 1: You can't say that every outcome has positive probability when there are uncountably many. The important part of assumption 1, though, is that likelihood ratios are uniformly bounded and bounded away from zero, for any two actions, as we vary the signals. These are technical difficulties, and while there is some economic intuition to them (see problem 8), we will not pursue them here.

We should, however, discuss the technical difficulties that arise when actions or effort levels are chosen from some interval. To fix matters, we use the model of two possible outcomes, success and failure, and an action choice that bears the interpretation of "effort." Effort is chosen from an interval $[a_0, a_1]$ where effort is measured on the scale of disutility; i.e., $U(w, a) = u(w) - a$ for some function u satisfying assumption 2. If effort level a is chosen, then the probability of a successful outcome is $\pi(a)$, where $\pi(\cdot)$ is a strictly increasing function.

How are we to construct the subproblem analogous to (Cn) in this

^f Or, to be very precise, this is so in cases where the first mover has a rich enough strategy space so that she could, at vanishing cost, "enforce" the choice she wants.

16.4. Continuous actions: The first-order approach

case? If we want to discover the minimal cost of implementing a^* , we would seem to want to solve

$$\begin{aligned} & \text{minimize } (1 - \pi(a^*))v(x_f) + \pi(a^*)v(x_s) \\ & \text{subject to } (1 - \pi(a^*))x_f + \pi(a^*)x_s - a^* \geq u_0 \\ & \text{and } (1 - \pi(a^*))x_f + \pi(a^*)x_s - a^* \geq \\ & \quad (1 - \pi(a))x_f + \pi(a)x_s - a \text{ for all } a \in [a_0, a_1], \end{aligned}$$

where the decision variables x_f and x_s are the utility wage levels contingent on failure and success, respectively.

The last "constraint," which is really an infinite number of constraints, is a killer to handle analytically. So in much of the early literature the following reasoning was employed. The last "constraint" can be recast as: The function

$$U(a) = (1 - \pi(a))x_f + \pi(a)x_s - a$$

should be maximized at $a = a^*$. The maximand of this function, assuming that it doesn't fall at one of the endpoints of $[a_0, a_1]$, can be found from the first-order condition $U'(a) = 0$ or

$$\pi'(a)[x_s - x_f] = 1.$$

So simply substitute this first-order condition into the subproblem formulation in place of the last constraint. That is, rewrite the problem of implementing a^* at minimal expected cost as

$$\begin{aligned} & \text{minimize } (1 - \pi(a^*))v(x_f) + \pi(a^*)v(x_s) \\ & \text{subject to } (1 - \pi(a^*))x_f + \pi(a^*)x_s - a^* \geq u_0 \\ & \text{and } \pi'(a^*)[x_s - x_f] = 1. \end{aligned}$$

Fixing a^* , this gives us two equations in two unknowns (assuming both constraints bind), and we are virtually home free.

Or are we? Is it legitimate to replace the infinite set of relative-incentive constraints with a single "first-order constraint?" In general, it is not. But one can give conditions under which this approach does work. Suppose assumption 5 holds. In this context, this is just a statement that $\pi(\cdot)$ is a concave function. Then for any nondecreasing wage-incentive scheme (that is, any scheme that pays at least as much for success as for

failure), we know from earlier analysis that the agent's expected utility function $U(a)$ as defined above is concave. If $\pi(\cdot)$ is assumed to be differentiable, then $U(\cdot)$ is also differentiable. And we then know that solution of the first-order condition is necessary and sufficient for an optimum, at least off the boundary. Moreover, in this two-outcome case, it is easy to argue from first principles that the optimal incentive-wage system is non-decreasing. So we know indeed that this approach is valid in this case if assumption 5 holds. But if assumption 5 doesn't hold, then one doesn't always obtain the optimal solution by this process. In particular, a scheme designed to satisfy the participation and first-order constraints may not satisfy all the relative-incentive constraints; i.e., it may not even be a feasible solution to the original problem.

When you read in the literature about the first-order approach to the principal-agent problem and about problems in that approach, it is this to which reference is being made.

Let us harvest at least one simple result from this first-order approach. Assume that $\pi(\cdot)$ is concave, so the approach is valid, and assume that $\pi(\cdot)$ is continuously differentiable. Consider the problem of implementing effort level a . The first-order constraint is

$$\pi'(a)[x_s - x_f] = 1 \quad \text{or} \quad x_s - x_f = \frac{1}{\pi'(a)}.$$

Substituting this into the participation constraint, we obtain the solution

$$x_f(a) = u_0 + a - \frac{\pi(a)}{\pi'(a)} \quad \text{and} \quad x_s(a) = u_0 + a + \frac{1 - \pi(a)}{\pi'(a)}.$$

Hence the expected level of wages that must be paid to implement a is

$$C(a) = (1 - \pi(a))v(x_f(a)) + \pi(a)v(x_s(a)).$$

Note well: This doesn't apply to $C(a_0)$; by the (I hope by now) obvious argument, to implement the lowest level of effort the principal should pay constant wages of $v(u_0 + a_0)$, which gives $C(a_0) = v(u_0 + a_0)$. We have:

Proposition 5. Assume that u is strictly concave and that assumptions 1 through 5 all hold.

(a) The function $x_f(a)$ is nonincreasing in a .

(b) The function $x_s(a)$ is nondecreasing in a .

(c) The function $C(a)$ is increasing in a .

(d) All of these functions are continuous for $a > a_0$. At $a = a_0$ they jump as follows: x_f jumps down; x_s and C jump up.

We leave the proof to you, with the following hints. For (a) and (b), take the derivatives of these functions, being careful about the value a_0 . For (c), use parts (a) and (b), the fact that $(1 - \pi(a))x_f(a) + \pi(a)x_s(a) = u_0 + a$, and the convexity of v . (Draw a picture of v , and locate the points $x_s(a)$ and $x_f(a)$ for two values of a .) Continuity of all these functions for $a > a_0$ follows from the fact that they are made up of continuous functions. The jump discontinuities rely on the fact that $\pi'(a_0) > 0$, which follows from concavity of π and assumption 3.

An economically interesting part of this lemma is the upward jump in C at a_0 . Since $B(a)$ is a continuous function of a , we will not tend to find optimal incentive schemes that implement a "small" amount of effort relative to the lowest possible level. A discrete cost must be paid to get the agent to do anything above the minimum, and that discrete cost must be "covered" by a discrete increase in benefits, which in turn requires a discrete step-up in the amount of effort chosen by the agent.

Since you will see it in many papers on the subject, let us close here with the formulation in the case where the set of outcomes is an interval $[s_0, s_1]$ from R . (To follow precisely what comes next, you need to know about Lagrangians for problems with infinitely many decision variables. But even if you don't know about such rarified optimization theory, you may get the basic idea by reading on.) We assume that as a function of a the distribution of outcomes is given by the cumulative distribution function $\Phi(s; a)$, which is assumed to have a continuous density function $\phi(s; a)$. We write $x(s)$ for the wage-utility if the outcome is s . Then for a fixed incentive scheme given by $x(\cdot)$, the utility to the agent of expending effort a is

$$\int_{s_0}^{s_1} x(s)\phi(s; a)ds - a,$$

so the first-order condition that a^* is optimal is

$$\int_{s_0}^{s_1} x(s)\phi_a(s; a)ds = 1,$$

where $\phi_a(s; a)$ means the partial derivative of $\phi(s, a)$ with respect to a . The subproblem with this first-order constraint put in place of the relative

incentive constraints is

$$\begin{aligned} & \text{minimize } \int_{s_0}^{s_1} v(x(s))\phi(s; a)ds \\ & \text{subject to } \int_{s_0}^{s_1} x(s)\phi(s; a)ds - a \geq u_0 \\ & \text{and } \int_{s_0}^{s_1} x(s)\phi_a(s; a)ds = 1. \end{aligned}$$

Letting λ be the multiplier on the participation constraint and η the multiplier on the first-order constraint, the entire problem has "first-order condition"

$$v'(x(s)) = \lambda + \eta \left(\frac{\phi_a(s; a)}{\phi(s; a)} \right).$$

(When and if you see this in other places, you may see the left-hand side written as $1/u'(w(s))$, where the decision variable is the wages paid if the gross profit level is s . Of course, by the formula for the derivative of the inverse of a function, these come to the same thing.)

Our remarks about when the first-order approach is valid in the case of two signals carries over to this situation. If the appropriate analogue of assumption 5 holds, then the agent's objective function is concave for any nondecreasing wage-incentive scheme, and the agent's optimal response is characterized by the first-order condition. Moreover, in the spirit of our analysis in the previous section, one can show that the optimal wage-incentive scheme will be nondecreasing if, in addition to the analogue of assumption 5, the appropriate analogue of the monotone likelihood ratio property holds.

From this "first-order equation" one can obtain a number of interesting results. In particular, watch out for the so-called Holmstrom informativeness condition (Holmstrom, 1979; Shavell, 1979), which concerns when information will be used in an optimal incentive scheme.

16.5. Bibliographic notes and variations

The story developed in sections 16.2, 3, and 4 is the basic *principal-agent* problem, the starting point of the literature on incentives. We have dealt primarily with the problem in the context of labor contracts, where the notion is to induce an effort-averse worker to choose a desired level of effort. But other contexts have been explored. These ideas appear in the realm of public finance; one issue that is explored there is the design of an income tax that maximizes net tax revenue (or maximizes some particular social welfare functional), recognizing that workers may work less if the marginal tax rate is high. These ideas also appear in the realm of insurance

markets where moral hazard is a problem. In each of these cases, there are reasons why it is not desirable to have the "agent" fully internalize the consequences of his actions: In the context explored in this chapter and in the context of hazard insurance, to share risk efficiently; and to promote equity or provide public goods in the context of public finance. The hallmark of these models is the trade-off between having the agent bear fully the consequences of his actions and obtaining the "other" sort of desirable end.

J. Mirrlees (1975) has made many of the significant contributions to the development of this theory, but his contributions are largely unpublished. The treatment given here relies heavily on Grossman and Hart (1983), and you would do well to see also Holmstrom (1979), Shavell (1979), Rogerson (1985), and Jewitt (1988) for related developments of the basic model.

A very large number of variations and elaborations on the basic model have been analyzed, some of which lead to results that cast quite a different light on the entire picture. We cannot hope here to give a complete survey, but we can offer a quick overview of a few interesting variations. (The references that are supplied are convenient and very far from exhaustive.) You may think of these variations as following (or combining) four basic lines: Incentive problems where the action choice is not interpreted as simple effort; problems where more than one agent and or principal is present; problems where time plays a significant role (the principal and agent deal not once but repeatedly and, perhaps, interactively); and problems concerning incentives when "agents" interact in imperfectly competitive environments.

Incentives to take appropriate decisions

The effort-aversion story we have given is not all that palatable in many contexts. When we think of providing incentives for managers, for example, it is not that the managers are necessarily effort averse, but that they tend to bend their efforts in directions that the principal might not particularly desire. (For the perspective of a sociologist on this, see J. Baron [1988].) This has led researchers to look into different sorts of formulations of the moral hazard problem, where incentives are designed to motivate agents to work according to the desires of the principal. Included here are studies of situations in which the agent will before taking an action receive some information that will be useful in making the decision. (Cases where the agent *already* possesses the information prior to the stage where he must accept or reject the contract are another variation, but this blends together material from this chapter and the next two.)

The general "technology" of section 16.3 can be adapted to analyze such situations. In essence, one specializes not with the story of effort aversion that we analyzed in some detail but with some other special structure.

There is a large variety of models of this sort in the literature, and any list is bound to be woefully incomplete. Moreover, many of these variations are set up to interact with factors that arise in multiperson or multiperiod situations to be discussed in a moment. A short list would include: Incentives for managers to invest capital fruitfully instead of in perks (Grossman and Hart, 1982); incentives to invest in human capital (Holmstrom and Ricart i Costa, 1986); incentives to obtain information about and invest in risky assets (Lambert, 1986); incentives issues in defense contracting (Baron and Besanko, 1987); incentives issues in regulated industries (Baron, 1989); incentives to obtain and use information in general (Demski and Sappington, 1987).

Multiple agents and multiple principals

Suppose that the principal in section 16.2 had not one but two salesmen going after different sales. Suppose that the reason one can't be sure what size sale will be made given the level of effort is that a competitive product might be better. Insofar as this competitive product affects the outcome to both salesmen, one would expect the outcomes that they achieve to be correlated. If one makes a large sale, then it is more likely that the competitive product is not so good. This increases the odds that the other will make a large sale, if he puts in the effort. In such a case, the principal might do well to structure the incentives so that the payment to one salesman depends on the outcome achieved by the other. For example, the principal might give them base wages, commissions on their own levels of sales, and a special bonus to the top salesman of the month/year. Or the principal might use an incentive scheme that is purely ordinal: So much for the agent who does the best, so much for the agent who comes in second, and so on. The literature on *tournaments* is about this sort of comparative incentives system.

More generally, when the random factors that affect the measured "performance" of one agent are correlated with factors that affect the measured performance of others, it can make sense to engage in relative performance evaluation. Analyses include Baiman and Demski (1980), Green and Stokey (1983), Holmstrom (1982b), Lazear and Rosen (1981), Mookherjee (1984), and Nalebuff and Stiglitz (1983). An issue that arises is whether agents will play the desired equilibrium when there is relative performance

evaluation; see Demski and Sappington (1984) and Ma, Moore, and Turnbull (1988).⁶

For an example along these lines, see problem 3.

In the literature on tournaments and relative performance evaluation, it is usually assumed that each agent has a project on which he works, and for each project there is some (imperfect) indication of how hard the agent worked. A different situation arises when there is group production; a number of individuals all contribute to a single project, with a single signal produced of how hard they (collectively) worked. This leads in turn to questions concerning internal versus external monitoring and incentives and to questions about task design and work-group composition. (Do you put all the hardworkers in one group or do you spread them among several groups?) For models along these lines, see Holmstrom (1982b), McAfee and McMillan (1986), and Lazear (1989). In a variation each agent has his own project, but can devote some time and effort to helping fellow agents. The principal in such a case may wish to provide incentives that promote helping efforts. See, for example, Itoh (1988).

A different sort of multiperson variation concerns situations where one agent works simultaneously for many different principals. Think, for example, of a consultant who is kept on retainer or a salesman who sells products for many different companies. In this case there would be competition among the principals to get the agent to devote relatively more time and effort to their interests at the expense of the interests of the other principals. See Bernheim and Whinston (1986).

Multiperiod incentives

When the relationship between the principal and agent is of longer duration, then opportunities may appear to improve the incentives given to agents. Radner (1985) suggests that with many independent signals of how hard an agent is working one can use the law of large numbers to provide first-best incentives. Fudenberg, Holmstrom, and Milgrom (1988) provide further analysis of this and argue that the result is due more to the ability to spread risk through time in a multiperiod setting.

The models we analyzed suggested that optimal incentive schemes will in general be very complex, depending on the very fine structure of the environment. This is not a prediction that is verified empirically; incentive schemes in practice are usually quite simple. One explanation might be

⁶ If the question here seems cryptic, wait until chapter 18.

that our models assume the principal has very precise information about the (distribution of) consequences of the agent's actions; if the details of the situation are somewhat hazy (which, presumably, one could model with incomplete information), then incentive schemes that are more robust to changes in the fine structure of the model might be more likely to be employed. A second explanation is offered by Holmstrom and Milgrom (1987). Within the context of a multiperiod, stationary environment, they show that optimal incentives will be "linear" in an appropriate sense. In essence, the idea is that the agent's ability to change his actions through time as consequences are realized gives him a great deal of control over the overall outcome. This, in turn, gives the agent the ability to manipulate any incentive scheme that depends on the fine details of an outcome; to avoid this, the principal chooses an incentive scheme that depends coarsely and simply on the overall outcome.

In the models so far discussed, the characteristics of the agent and the various tasks were held fixed and were assumed to be known to both the principal and the agent. But in many applications of interest, the difficulty of a particular task is not known a priori. In multiperiod contexts this complication multiplies. The principal observes what outcomes the agent obtains in early periods to try to gauge the difficulty of the task. If the agent does well early on, the principal will come to believe that the task is relatively easy, and she will, therefore, lower the incentive wages she is willing to pay. An agent who knows this is going to happen will in response tend to underperform early on to make the principal think that the job is relatively difficult. The principal, in equilibrium, is not deceived by this, but still in equilibrium the agent will hold back early on, so as not to convince the principal that the task is very easy. In general, the principal would benefit if, ex ante, she could commit to an incentive system that will not change as results unfold. But when she cannot so commit, this sort of *ratchet effect* complicates both the incentive system she puts in place and the agent's response to that system. For discussion and analysis, see Gibbons (1987) and Laffont and Tirole (1988).

The other side of this coin finds the principal unsure, a priori, how much ability the agent has. We imagine that the measured outcome of an agent compounds the agent's effort and some exogenous noise, as in this chapter, but also the agent's ability. Moreover, we imagine that the principal is willing to pay relatively more for a more able agent. (Think of there being a competitive market for labor services, although see the next paragraph.) Then insofar as a high level of output from an agent early on is taken as (partial) evidence that the agent is more able, the agent has a relatively greater personal incentive to work hard earlier; the principal, in

response, may have less need to provide for immediate direct incentives. An early model along these lines is provided by Holmstrom (1982a).

Continuing along these lines, the question how an agent's immediate output affects the evaluation of the agent and his subsequent opportunities and pay has been the subject of much analysis. The general question of career concerns and its effect on incentive structures is investigated by Holmstrom (1982a). Milgrom (1988) and Milgrom and Roberts (1989) explore the incentives for workers to exert influence on their superiors and the consequences (in terms of optimal incentive structures) for organizational design. Another subject of interest is the existence of so-called internal labor markets, where employees of one firm rise within the firm; that is, workers join the firm primarily at low level "ports of entry" and then progress through the ranks. See Gibbons (1985) and MacLeod and Malcolmson (1988) for two analyses.

Incentives for agents who interact

Imagine a duopoly in which the firms compete in, say, Cournot fashion. Suppose that one firm is managed by its owner, but the second firm has a manager who is distinct from the owner. The owner of this second firm wishes to provide an incentive contract for her manager that will leave her (the principal) with as much net profit as possible. Beyond the sorts of issues that we considered, a new issue intrudes here: How will the incentive scheme employed for the manager of the second firm affect the production decisions of the owner-manager of the first firm? If, to take a simple example, the owner of the second firm could give her manager a contract where he only made money if he chose the von Stackelberg level of output, and if the owner-manager of the first firm knows this, then the owner-manager of the first firm will not respond (in equilibrium) as aggressively as if the second firm also had an owner-manager.

In general, when one provides incentives to an agent who interacts with others in a gamelike situation, the incentives provided can affect both what the agent and others do. This can significantly complicate the problem of designing incentives. See problem 2 for an example, and see Fershtinan and Judd (1987) and Maksimovic (1986) for analyses.

Contextual applications

In the literatures of academic accounting, finance, operations, and marketing one can find many interesting applications of incentive theory. These applications are especially interesting because the models are often created with very specific institutions in mind. Some of the papers cited above are drawn from these literatures, and I will not attempt here to list

others. But in looking for applications and variations, one should consult not only the economics journals but also journals such as the *Journal of Accounting Research*. Your attention is called in particular to one contextual application — the effect of the tax code on incentives, as studied in Scholes and Wolfson (forthcoming).

It would be easy to go on for pages more, listing interesting variations on the basic story given in this chapter. And a book of quite substantial length could be written giving details. Not having that much space, we suspend discussion here, although you will quickly realize that most of the remainder of the book deals in varying levels of formality with the problem of incentives to do one thing or another.

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16.6. Problems

- 1. To test your understanding of the model in section 16.2 consider the following reparameterization. The reservation utility level of the agent is only 3. A high level of effort gives $a = 3$, while a low level gives $a = 0$. (Thus, if we can contract contingent on the level of effort, we would have to pay \$9 for a low level of effort and \$36 for high.) A high level of effort gives no sale with probability .2, a \$100 sale with probability .4, and a \$400 sale with probability .4. For a low level of effort, the three probabilities are .4, .4, and .2.
- (a) What is the optimal way to induce the agent to put in a low level of effort, if you can contract contingent on the size of the sale (only)?
- (b) What is the optimal way to induce the agent to put in a high level of effort?

- (c) What is the optimal contract (if you are a risk neutral principal) to offer this agent?

■ 2. The point of this problem is "incentives meet duopoly theory." Consider a simple duopoly problem, with two firms that are identical except in one important sense. The two produce an undifferentiated good, the (total) demand for which is given by the demand curve $P = A - X$, where P is price and X is total quantity produced. The two compete as Cournot competitors; each names a quantity supplied, simultaneously and independently of the other, and market price is set so that demand equals the sum of the two supplies. Both firms have zero marginal costs of production. (So far, completely standard.)

One firm is owner managed. The quantity decision for this firm is made by the owner, who retains any profits. This owner is risk neutral.

The second firm is not managed by its owner. The owner of the second firm has hired a manager and has given this manager a contract wherein the compensation paid to the manager is a linear function of the profits the firm shows, π , and the quantity that the firm sells, x_2 . Specifically, this contract calls for the manager to be paid $\alpha\pi + \beta x_2$, where α and β are given constants. The owner retains any profits made by the firm, net of pay to the manager. Note well that the manager makes the quantity decision and not the owner, and the manager does so to maximize her compensation and not the net profits of the owner.

- (a) What, in this case (for given α and β), is the Cournot equilibrium in the market? What (as a function of α and β) are the profits of the first firm, the manager's compensation, and the net profits (to the owner) of the second firm?

(b) The owner of the second firm is interested in designing the "optimal" contract for her manager. The contract must, in equilibrium, provide the manager with a reservation level of income Y . Other than that, the owner can choose any linear contract she wishes; α and β are chosen to maximize her net (after compensation) profits. The owner-manager of the first firm and the manager of the second firm will behave in Cournot fashion after this contract is chosen. If you are worried about such things, the owner-manager of the first firm will know the contract terms under which the manager of the second firm operates. What is that optimal contract? (Hint: Draw pictures!)

- (c) Suppose now that both firms are managed by managers distinct from owners, both managers are paid according to contracts that have com-

pensation linear in profits and sales quantity, and both managers have a reservation level of compensation that must be met. The two owners simultaneously and independently set the terms for their managers' (respective) contracts. What will be the equilibrium in this game?

(d) We can imagine, in (c), many possible ways this game is played. Here are two extremes.

In the first, the two owners simultaneously name contract terms for their managers. Those terms are named publicly, so each manager sees the terms offered to the other. The two managers then decide whether to take the contract offered. They do so if, in the subsequent round of quantity setting, they will get their reservation level of compensation. Then the managers set quantities. If only one manager signs, the other firm (the one without a manager) is out of the market, and the one with a manager acts as a (compensation-maximizing) monopolist.

In the second extensive form, the two owners simultaneously name contract terms, and the two managers either sign or not and choose quantities without knowing what is going on at the other firm. If the manager at a firm doesn't sign, then that firm cannot produce.

We're only looking for (subgame) perfect equilibria. Then does the formulation matter? In particular, which (if either) of these formulations have you assumed in your answer to part (c)?

■ 3. In this problem, we will deal with incentives for multiple agents. You are a manager in charge of two workers. Each worker is asked to perform a particular task, and each can either work hard at the task or loaf. The value to you of each task will be either \$10 or \$0. The value is determined in part by how hard the worker works, but there are also some random factors not within the control of the worker. The following data apply:

If the worker works hard, there is a .7 probability that the value to you will be \$10 and a .3 chance that the value will be \$0. If the worker loafs, these probabilities reverse, so that there is a .3 chance of a \$10 outcome and a .7 chance of a \$0 outcome.

Moreover, there is correlation between the outcomes of the two tasks:

- If both workers work hard, then there is .6 joint probability that each will produce a \$10 outcome
- If both loaf, there is a .2 joint probability that each will produce a \$10 outcome

- If one loafs and the other works, there is a .25 joint probability that each will produce a \$10 outcome

(Start by putting together joint probability tables from the data above.)

Each worker has a utility function that depends on the amount of money received and the amount of effort expended. The form of the utility function is

$$U(w, a) = \sqrt{w} - a,$$

where w is the amount of money received, and $a = 0$ for loafing and $a = .8$ for working hard. The two workers have the same utility function, and each has a reservation utility level of 1. The worker's expected utility must be 1 to get the worker to take the job.

You as the employer are risk neutral. You seek to maximize your net profit from these two workers.

(a) Suppose that you could at zero cost monitor the efforts of the two workers. Would you want to make them work hard, or is it better to have them loaf? What payments would you offer them?

(b) Suppose that you cannot monitor the efforts of the two workers. You can, however, sign a contract of the form: I agree to pay \$X if the task you perform has value \$10 to me, and \$Y if the task has value \$0. Subject to the constraints that the worker must be willing to sign, and the worker, seeing what is being paid, will choose how hard to work, what is the best contract of this form for you? (The worker will resolve all ties in his utility in your favor.)

Now suppose that you cannot monitor the two, but you can offer them more complex contracts, of the following form: I agree to pay \$X if both of you produce a \$10 outcome for me, \$Y if you produce a \$10 outcome and your fellow worker produces \$0, \$Z if you produce \$0 and your fellow worker produces a \$10 outcome, and \$W if you both produce \$0.

(c) What contracts of this form are optimal if you want to induce both workers to loaf?

(d) Suppose that you wish to offer contracts that will induce both to work hard. Imagine that this means you must meet the following constraints: Each worker must be induced to take the contract; each worker, assuming that his fellow worker is going to work hard, must be induced to work hard. What is the optimal contract to offer in this case?

(e) You are worried that the contract offered in (d) might induce the following behavior: Each worker will loaf. You would feel less worried about this possibility if you knew that each one, given the other was loafing, would prefer to work hard. If you add this constraint to part (d), is the old solution still valid? If not, what is the new solution?

(f) Can you do better to induce one worker to loaf and the other to work hard? How well do you do in this case?

■ 4. I am trying to decide on the scale of investment that I will make in a particular project. The returns on my investment will depend on the scale of the investment and one environmental variable that will either be S or F . Right now, I think that the probability of S is .7. But out there is someone who could, with some effort, refine that assessment. This individual can run a test that will give one of two outcomes s or f , each with probability .5. If the test gives outcome s , then the posterior probability of S is .9, while if the outcome of the test is f , then the posterior probability of S is .5.

If I hire this individual, I can't observe whether she has run the test or, if she has, what outcome she actually got. She can report to me the outcome, but I will have to take her word for the fact that she ran the test and reported truthfully what the test result was. If she is indifferent between two reports or between running the test or not, she will resolve ties in the way that is most favorable to me.

I can pay this individual depending on her report and on the ultimate outcome S or F of the environmental variable.

Her von Neumann-Morgenstern utility function is given by

$$\sqrt{\text{Dollar payment to her} - f(\text{effort})},$$

where $f(\text{no effort}) = 0$ and $f(\text{run the test}) = f > 0$. She has a reservation expected utility level of f' .

My utility takes the form

$$\begin{aligned} & \text{Expected Gross Benefits(my investment decision, true state)} \\ & - \text{Expected payment to her.} \end{aligned}$$

I need some help in deciding whether to hire this person and, if I hire her, how to structure the payments I make to her. Help me. Write a (maximum) two-page memo to me outlining what I should do and why.

Please note that you don't have enough data to tell me whether to hire, or what contract to give her exactly. All that is desired is an outline of how I should go about solving the problem (justifying your suggested method of solution).

■ 5. Consider an individual who owns a warehouse that is subject to a fire danger. Suppose that the warehouse, if it burns at all, suffers damage that has a uniform distribution over the range from \$10,000 to \$190,000. The owner of this warehouse can take precautions against the event of fire — for simplicity, we imagine that the owner can either "take care" or "be negligent." The owner's decision whether to take care or be negligent affects the probability whether there is a fire at all: If she takes care, the probability a fire occurs π_1 , while if she is negligent, the probability of a fire is $\pi_2 > \pi_1$. The extent of damage is independent of the action the owner takes, if there is a fire.

The owner's preferences are described by a von Neumann-Morgenstern utility function U that depends on the level of damages incurred in a fire (if any) and the actions she takes to prevent a fire. We write her utility function as $U(K - L - E)$ where K is some constant, L is the size of her loss, if any, and $E = 0$ if she is negligent and $E = \$10,000$ if she takes care. Note that if there is no fire, her utility will be $U(K - E)$. Assume that U is concave.

This individual has the opportunity to insure against her loss by fire. A typical insurance contract will take the following form: The owner pays a premium P up front; if the owner sustains a loss of size L , then the contract specifies an amount $\alpha(L)$ that is returned to the owner. In this case, the owner's utility is $U(K - L + \alpha(L) - E - P)$ if she has a fire with loss at level L and $U(K - E - P)$ if she has no fire. (We implicitly assume that $\alpha(0) = 0$.)

The insurance companies with which she might deal are all regulated by government authorities, and they are required to offer insurance at premiums that on average will net zero profit. That is, the premium P must be set to equal the expected payout on the insurance policy. (If you want to make the problem more realistic, you can change this. Assume instead that the insurance company is regulated so that some fixed small percentage of the premium it charges is taken as profit. This will, however, make some of the answers obtained below less clean than in the formulation we gave.) The owner can deal with at most one of these insurance companies.

(a) Suppose that insurance companies are compelled to offer complete

coverage: $\alpha(L) = L$. What will be the results in terms of the insurance policy that is offered and the level of care the warehouse owner will take?

(b) Suppose insurance companies can offer insurance that insures only a portion of the loss incurred: $\alpha(L) = \gamma L$ for some $\gamma < 1$. Show by means of a numerical example that this might make the warehouse owner better off than if the insurance companies were compelled to offer full coverage.

(c) Suppose insurance companies can offer insurance that insures the full loss but with a deductible: $\alpha(L) = \max\{L - D, 0\}$ for some fixed deductible D . Show by means of a numerical example that this might make the warehouse owner better off than if the insurance companies were compelled to offer full coverage.

(d) Consider the problem of designing an insurance policy of the sort given above (where there is a fixed premium and a payment given back to the owner in the event of a fire in an amount that depends on the losses sustained) that is "optimal" in the sense that it makes the warehouse owner as well off as possible, given the regulatory constraint that the insurance company must break even on average. How would you go about finding this optimal policy? Can you say anything about the function $\alpha(L)$ at the optimal policy? For example, does $\alpha(L)$ rise with L ?

(e) Suppose that the size of the loss sustained by the warehouse owner, given there is a fire, depends on the care she did or didn't take. More specifically, suppose that she is more likely to sustain a larger loss, given she has a fire at all, if she is negligent than if she takes due care. How does this affect your answers to part (d)?

■ 6. Consider the basic problem of section 16.3, but in a situation where the agent's utility function $U(w, a)$ is not necessarily additively separable. We assume that for each fixed a , $w \rightarrow U(w, a)$ is continuous, strictly increasing, and concave. Reformulate the method of solution of the basic problem in this case. (About the only thing to do here is decide on what you will make your decision variables and then write things down. Don't look for great mysteries or complications.)

The next three problems concern assumption 1. Problem 7 is relatively easy, and everyone should try it. Problem 8 is a bit more difficult. Problem 9 involves a fairly high level of skill at mathematical analysis.

■ 7. Suppose that we have a two-action, two-outcome principal-agent problem. The outcomes are denoted s_1 and s_2 and represent gross profits to the principal with $s_2 > s_1$. The two actions are effort levels, denoted a_1

and a_2 , with $a_1 < a_2$. The agent's utility function is of the form $U(w, a) = -e^{-w} - a$, where a denotes effort and w is the wage payment. The first effort level results in outcome s_1 with probability .1 and s_2 with probability .9. The second effort level results in outcome s_2 with certainty. Show that it is possible, with a contract that depends on gross profit levels only, to implement the action choice a_2 at an expected wage payment that is arbitrarily close to the cost of implementing a_2 in a contract that can specify the action choice. (Note well: We don't constrain the wage payments to be nonnegative.)

■ 8. Suppose that we have a two-action, principal-agent problem where the range of possible outcomes is the entire real line R . As in problem 7, outcomes are gross profit levels. The agent's utility function is precisely as in problem 7. If the agent takes action a_1 , the level of gross profits is Normally distributed with mean μ_1 and standard deviation $\sigma > 0$, whereas action a_2 gives gross profits that are Normally distributed with mean μ_2 and standard deviation σ for $\mu_2 > \mu_1$. Show that it is possible with a contract that depends on only gross profit levels to implement the action choice a_2 at an expected wage payment that is arbitrarily close to the cost of implementing a_2 in a contract that can specify the action choice.

■ 9. Prove that if assumptions 1 and 2 hold and if there is any set of decision variables (x_m) that satisfies the constraints of the problem (C_n) for given n , then the problem (C_n) has a (finite) solution. (Hints: To prove this, consider separately the case where u is linear and where it is not. When u is linear, think of the problem as a linear programming problem. The only way we could fail to find a finite solution is if we found an infinite ray along which the objective function decreased indefinitely. But from the participation constraint, we know that $\sum_{m=1}^M \pi_{nm} v(x_m)$ is bounded below by $a_m + u_0$. If, on the other hand, u is not linear, but concave, you can show that any unbounded sequence of solutions (x_m^k) of the participation constraint (where k indexes the sequence) must have an objective function value that diverges to $+\infty$. Hence you can restrict attention to bounded sequences of solutions of the constraints.)

■ 10. Prove that if assumption 3 holds, then $\sum_{m=1}^M \pi_{nm} s_m$ is increasing in n . (Hint: Read the proof of the first lemma after assumption 5. The key step is the interchange of the order of addition right near the start. If you understand that step, you'll see how to prove this.)

■ 11. Consider the second part of the example in the subsection of section 16.3 on the two-outcome case. What would happen if a_3 turned out better

than a_2 at these wages? (Can you change the numbers so that this is so?) What would $C(a_2)$ be in that case?

■ 12. Prove proposition 5.

■ 13. Return to the finite action, finite outcome formulation of section 16.3 in the specialization where a is effort and the signals s are levels of gross profits. Assume that assumptions 1, 2, and 4 all hold and that u is strictly concave. An alternative to assumption 5 is: There are two probability distributions $(\pi_1^1, \dots, \pi_M^1)$ and $(\pi_1^2, \dots, \pi_M^2)$ such that for every action a_n there is a number $\alpha_n \in [0, 1]$ with $\pi_{nm} = \alpha_n \pi_m^1 + (1 - \alpha_n) \pi_m^2$. (Assume that α_n is increasing in n . What conditions must hold between π^1 and π^2 so that assumption 4 is guaranteed?) Show that this alternative with assumptions 1, 2, and 4, is sufficient to show that the optimal wage-incentive scheme is nondecreasing in gross profits. (Hint: Do *not* try to prove that the only binding relative incentive constraints will be those of index lower than the effort level that is being implemented.)

chapter seventeen

Adverse selection and market signaling

Imagine an economy in which the currency consists of gold coins. The holder of a coin is able to shave a bit of gold from it in a way that is undetectable without careful measurement; the gold so obtained can then be used to produce new coins. Imagine that some of the coins have been shaved in this fashion, while others have not. Then someone taking a coin in trade for goods will assess positive probability that the coin being given her has been shaved, and thus less will be given for it than if it was certain not to be shaved. The holder of an unshaved coin will therefore withhold the coin from trade; only shaved coins will circulate. This unhappy situation is known as Gresham's law — bad money drives out good.

17.1. Akerlof's model of lemons

Gresham's law has had more recent expression in Akerlof (1970). (See problem 12 after you finish this section, however.) In Akerlof's context, Gresham's law is rephrased as "Bad used cars drive out good." It works as follows.

Suppose there are two types of used cars: peaches and lemons. A peach, if it is known to be a peach, is worth \$3,000 to a buyer and \$2,500 to a seller. (We will assume the supply of cars is fixed and the supply of possible buyers is infinite, so that the equilibrium price in the peach market will be \$3,000.) A lemon, on the other hand, is worth \$2,000 to a buyer and \$1,000 to a seller. There are twice as many lemons as peaches.

If buyers and sellers both had the ability to look at a car and see whether it was a peach or a lemon, there would be no problem: Peaches would sell for \$3,000 and lemons for \$2,000.

Or if neither buyer nor seller knew whether a particular car was a peach or a lemon, we would have no problem (at least, assuming risk neutrality, which we will to avoid complications): A seller, thinking she

has a peach with probability $1/3$ and a lemon with probability $2/3$, has a car that (in expectation) is worth \$1,500. A buyer, thinking that the car might be a peach with probability $1/3$ and a lemon with probability $2/3$, thinks that the car is worth on average \$2,333.33. Assuming an inelastic supply of cars and perfectly elastic demand, the market clears at \$2,333.33.

Unhappily, it isn't like this with used cars. The seller, having lived with the car for quite a while, knows whether it is a peach or a lemon. Buyers typically can't tell. If we make the extreme assumption that buyers can't tell at all, then the peach market breaks down. To see this, begin by assuming that cars are offered for sale at any price above \$1,000. All the lemons will be offered for sale. But only if the price is above \$2,500 will any peaches appear on the market. Hence at prices below \$2,500 and above \$1,000, rational buyers will assume that the car must be a lemon. Why else would the seller be selling? Given this, the buyers conclude that the car is worth only \$2,000. And at prices above \$2,500, the car has a $2/3$ chance of being a lemon, hence is worth \$2,333.33. There is no demand at prices above \$2,000, because (a) above \$2,333.33, there is no demand whatsoever — no buyer is willing to pay that much — and (b) below \$2,500 there is only demand starting at \$2,000, since buyers assume that they must be getting a lemon. So we get as equilibrium: Only lemons are put on the market, at a price of \$2,000. Further gains from trade are theoretically possible (between the owners of peaches and buyers), but these gains cannot in fact be realized, because buyers can't be sure that they aren't getting a lemon.

Note that if there were two peaches to every lemon, the story wouldn't be so grim. Then, as long as cars reach the market in these proportions, a buyer is willing to pay \$2,666.67. And that is enough to induce owners of peaches to sell; we get the market clearing at this price, with all the cars for sale. (What does the demand curve look like in this case? What does the supply curve look like? Are there other market clearing prices? See problem 1.) Owners of peaches are not pleased about those lemon owners; without them, peach owners would be getting an extra \$333.33 for their peaches. But at least peaches can be sold.

This is a highly stylized example of Akerlof's market for lemons. It illustrates the problem of *adverse selection*. Assume a particular good comes in many different qualities. If in a transaction one side but not the other knows the quality in advance, the other side must worry that it will get an adverse selection out of the entire population. The classic example of this is in life/health insurance. If premiums are set at actuarially fair rates

for the population as a whole, insurance may be a bad deal for healthy people, who then will refuse to buy. Only the sick and dying will sign up. And premium rates must then be set to reflect this.

The problem noted above becomes worse the greater the number of qualities of cars and the smaller the "valuation gaps," the differences between what a car is worth to a buyer and a seller, assuming they have the same information. Imagine, for example, that the quality spectrum of used cars runs from real peaches, worth \$2,900 to sellers and \$3,000 to buyers, down to real lemons, worth \$1,900 to sellers and \$2,000 to buyers. Between the two extremes are cars of every quality level, always worth \$100 more to buyers than to sellers. Suppose that the distribution of quality levels is uniform between these two extremes. To be very specific, suppose there are 10,001 cars, one of which is worth \$1,900.00 to its owner and \$2,000.00 to buyers, a second worth \$1,900.10 to its owner and \$2,000.10 to buyers, and so on. What will be the equilibrium then? (Continue to assume inelastic supply and elastic demand at every level of quality.)

Let us draw the supply curve first. At price $p = \$1,900$ there is one car offered for sale; at $p = \$1,901$ there are 11 cars offered, and so on. At $p = \$2,900$, all 10,001 cars are offered for sale, and that is all that is offered at any higher price. If we smooth out supply over the small discrete bumps, we get the supply curve shown in figure 17.1(a).

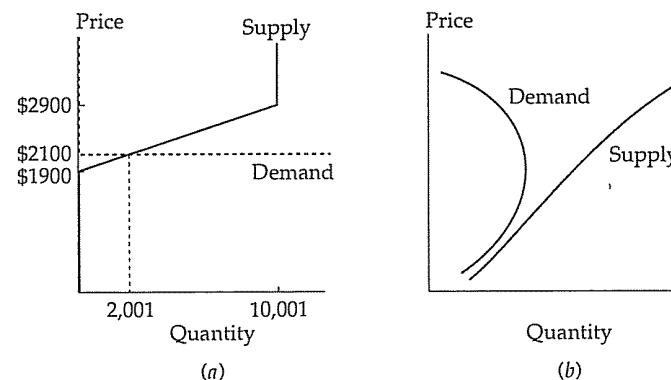


Figure 17.1. Supply and demand with adverse selection.

In (a), we have supply and demand for the second version of the lemons market. In (b), we depict the sad situation of upward sloping demand and a market that shuts down completely.

As for demand, at a price p between \$3,000 and \$2,000, buyers assume that only sellers who value their own cars at p or less are willing to sell. Hence a car being sold at price p has a quality level that makes it worth between \$2,000 and $$(p + 100)$ to buyers, with each value in this range equally likely. Therefore, the average car being sold is worth $$(2000 + p + 100)/2$. If p exceeds \$2,100, this average value is something less than p , and there is no demand. If p is less than \$2,100, this average value exceeds p , and there is infinite demand. At $p = \$2,100$ (plus or minus a penny) the average car offered for sale is worth \$2,100, and buyers are indifferent between buying and not. So we get the demand curve shown in figure 17.1(a) — no demand at prices exceeding \$2,100, perfectly elastic demand at $p = \$2,100$. The market equilibrium is where supply and demand cross, or $p = \$2,100$, at which only 2,001 of the 10,001 cars change hands. (And if the \$100 difference in valuations between buyers and sellers is \$50 instead, then the equilibrium price is $p = \$2,000$, and only 1,001 of the cars change hands.)

Let us do this one more time (just to pound it into your brain), at one level greater generality. We imagine that some item (a durable good, or some service) comes in one of N quality levels, $q^1, q^2, \dots, q^N$, arranged in ascending order. The supply of quality n depends on the price p and is given by an upward sloping supply function $s^n(p)$. Demand depends on price p and the average quality in the market, which by the assumptions we've made can be computed as

$$q^{\text{avg}}(p) = \frac{\sum_n q^n s^n(p)}{\sum_n s^n(p)}.$$

Let $D(p, q^{\text{avg}})$ be this demand function, which we will assume is decreasing in p and increasing in q^{avg} . Hence market demand at a price p is obtained, at least if buyers anticipate where supply will come from, as $D(p, q^{\text{avg}}(p))$. To find the slope of market demand we compute

$$\frac{dD}{dp} = \frac{\partial D}{\partial p} + \frac{\partial D}{\partial q} \times \frac{dq^{\text{avg}}(p)}{dp}.$$

The first term on the right-hand side is negative. The second is a positive term, times $dq^{\text{avg}}(p)/dp$, which may be positive; average quality supplied may be increasing in price. Hence it is possible if the second term is large enough that demand has positive slope for some levels of price. And, therefore, it is even possible that we get the sort of picture in figure 17.1(b),

where the only intersection between demand and supply is at zero; the market shuts down, even though gains from trade are possible.

17.2. Signaling quality

In spite of the lemons problem, used car markets and life/health insurance markets and all manner of markets subject to adverse selection do function. Why?¹ Often it is because the side to the transaction that has the superior information will do something that indicates the quality of the good being sold. With used cars, the seller may offer a partial warranty or may get the car checked by an independent mechanic.² In insurance, medical checkups are sometimes required. Another ploy in the realm of life insurance concerns "golden age" insurance, which is often marketed with the come-on that no one will be turned down. These policies often contain an important bit of fine print: "Benefits are greatly reduced for the first two years." That is, the insurance company is betting that if the buyer knows that he is going to die, it will probably happen quickly. The key to such signals is that the sellers of higher quality, or buyers in the case of insurance, to distinguish themselves from sellers of lower quality are willing to take actions the sellers of lower quality do not find worthwhile.

One can think of adverse selection as a special case of moral hazard and market signals as a special case of incentive schemes. We want the individual to self-identify as having a used car that is a lemon or a peach, as being gravely ill or not; and so on. But there is moral hazard in such self-identification; the seller of a used car cannot be trusted to represent honestly the quality of the car for sale unless provided with some incentive for doing so. The incentive could be relatively direct; the seller will be tossed in jail if he has egregiously misrepresented the quality of the car sold. Or it could be more indirect, as in the sort of market signals just described; the individual could be asked to give a six-month warranty if he says that the car is a peach. The point is that such an "incentive"

¹ We saw one possible answer in chapter 14 — reputation. But in this chapter, we consider cases where the informed party isn't able to make use of a reputation, say because this is a one-time transaction. We could also use reputation indirectly, through recourse to a market intermediary such as a reputable used car dealer. But that takes us to topics we will discuss in chapter 20.

² Since the buyer can't be sure how independent this mechanic is, in the used car market in the United States it is often the buyer who will provide the mechanic and the money for this. But the signal is still being provided by the seller, insofar as she allows the buyer's mechanic to investigate the car. By way of contrast, in other countries "inspection companies" are so well known and credible that it is customary for the seller to provide a report from one of them. Remember this example when we discuss trilateral governance in chapter 20.

contract" (I will pay you \$X if you don't give me a warranty and \$Y > \$X if you do) gives the party with (private information the incentive to self-identify/honestly, and the moral hazard problem of self-identification is defeated.

The two classic models of this in the literature are Spence's (1974) job market signaling model and Rothschild and Stiglitz's (1976) model of an insurance market. While cast in different settings, the two seem at first to be about the same problem. But their analyses come to different conclusions. We will use the Spence setting to illustrate their two analyses (and further variations):³

We imagine a population of workers, some of high innate ability, some of low. We will use t to denote the level of ability, with $t = 2$ for high ability and $t = 1$ for low. The numbers of high- and low-ability workers are equal. There are also firms that will (eventually) hire the workers; these firms operate in a competitive labor market and so will be making zero profits (when all the dust settles) out of their workers.

The key ingredient comes next: Before going to work, workers seek education that enhances their productivity. Specifically, each worker chooses a level of education e from some set, say $[0, 16]$, for the number of years in school.⁴ A worker of type t with education level e is worth precisely te to a firm. Note that for every level of education, a more able worker is worth twice as much as a less able worker. Firms, when they hire a worker, are unable to tell whether the worker is able or not. But they do get to see the worker's c.v., and they thereby learn how many years the worker went to school. Hence they can make wage offers contingent on the number of years that the worker went to school.

What do the workers want out of all this? They want higher wages, to be sure. But they also dislike education. And the less able dislike education even more than do the more able. Specifically, workers seek to maximize a utility function of wages and education, which we will write as $u_t(w, e)$, where w is the wage rate, e the education level, and t the worker's type. We assume that u_t is strictly increasing in w , strictly decreasing in e , and strictly concave. And we make the following very important assumption: Take any point (e, w) and consider the two indifference curves of high- and low-ability workers through that point. (Refer to figure 17.2, noting that the monotonicity and concavity assumptions we made concerning the functions u_t give us strictly convex indifference

³ In the problems, you will be given some leads on how these two stories are told in the case of insurance markets.

⁴ We allow any number between 0 and 16 to keep the pictures relatively simple. A treatment where education levels came only in discrete numbers would not be very different.

curves and the direction of increasing preference as depicted.) We assume that the indifference curve of the low-ability worker is always (more steeply) sloped (as in figure 17.2). That is, to compensate a worker for a given increase in education requires a greater (increase in wages for a low-ability worker than for a high-ability worker. Both sorts of worker dislike education, but low-ability workers dislike education relatively more (measured in terms of wage compensation). This assumption is crucial for what follows because it implies that a more able worker finds it (relatively) cheaper (in terms of utility) to obtain a higher level of education, which then can be used to distinguish more able workers from those of lower ability. The assumption is known informally as the *single-crossing property*.⁵ Utility functions of the form $u_t(w, e) = f(w) - k_t g(e)$, for f an increasing and concave function, g an increasing and strictly convex function, and the k_t positive constants with $k_2 < k_1$ constitute an example.

What would happen if the ability level of an employee were observable? Then a high-ability worker with education level e would be paid $2e$, and a low-ability worker with education level e' would be paid e' . As in figure 17.3, high- and low-ability workers will pick education levels that maximize their utility, given these wages.

But we are interested here in cases where ability level is not (directly) observable. Consider the following two stories: *job market signaling* and *then worker self-selection* (facing a menu of contracts).

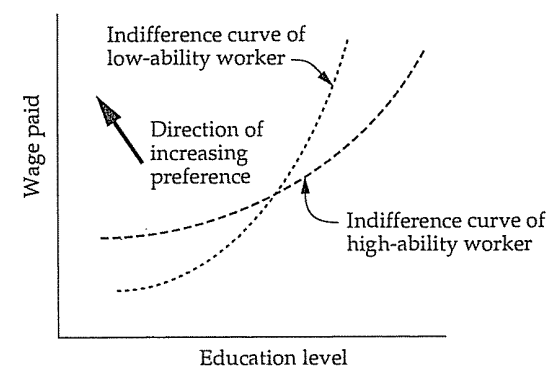


Figure 17.2. Worker indifference curves in wages-education level space.

⁵ It is usually formulated with a bit more precision than we have done here, and you will be given the chance to provide a more precise formulation in the problems.

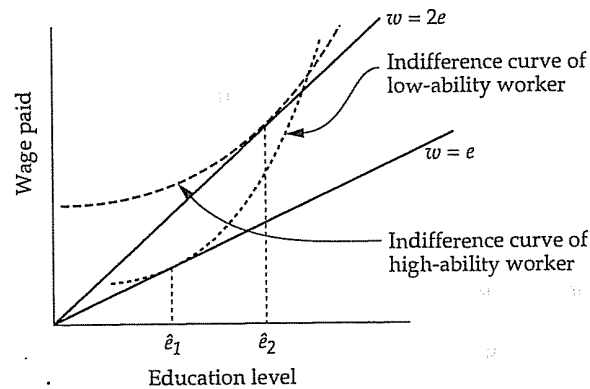


Figure 17.3. The full information equilibrium.

If ability is observable, high-ability workers will select education level $\hat{e}_2$ and be paid $2\hat{e}_2$, and low-ability workers will select $\hat{e}_1$ and be paid $\hat{e}_1$.

Spence's story: Job market signaling

Suppose that workers move first in the following sense. With no guarantees except that they expect the market mechanism to work when the time comes, workers choose how many years to go to school. They do so anticipating some wage function $w(e)$ that gives wages for every level of education. After they spend their time in school, they present themselves to a competitive labor market, and the firms in that market bid for their services.

Formally, an equilibrium in the sense of Spence consists of (a) an anticipated wage function $w(\cdot)$ that gives the wages $w(e)$ that workers anticipate will be paid to anyone obtaining this education level e , for every e , and (b) probability distributions π_t on the set of education levels for each type of worker $t = 1, 2$ such that the following two conditions are met:

- (1) For each type t of worker and education level e , $\pi_t(e) > 0$ only if $u_t(w(e), e)$ achieves the maximum value of $u_t(w(e'), e')$ over all e' .
- (2) For each education level e such that $\pi_1(e) + \pi_2(e) > 0$,

$$w(e) = \frac{.5\pi_1(e)}{.5\pi_1(e) + .5\pi_2(e)}e + \frac{.5\pi_2(e)}{.5\pi_1(e) + .5\pi_2(e)}2e. \quad (\clubsuit)$$

This has the following interpretation: $\pi_t(e)$ gives the proportion of type t workers who select education level e in equilibrium.^a Then condition (1) is a *self-selection condition*: Based on the wages that workers anticipate, they only select education levels that with the corresponding wage levels maximize their utility. And condition (2) says that at education levels picked in equilibrium the wages paid are appropriate in a competitive labor market for a worker offering that education level. This comes about because the two fractions on the right-hand side of $(\clubsuit)$ are the conditional probabilities that the worker is of low and high ability, respectively, obtained by Bayes rule.⁶ Hence the right-hand side of $(\clubsuit)$ is the conditional expected value of a worker presenting education level e , and (it is assumed) competition among firms pushes the worker's wage to that level.

In the definition of an equilibrium just given, we allow workers of a given type to choose more than one level of education. But we will restrict attention here to equilibria in which all the workers of a given type choose the same level of education:

In the first type of equilibrium, workers of the two types are separated; this is called a *separating equilibrium*. All the workers of type $t = 1$ choose an education level e_1 , and all the workers of type $t = 2$ choose an education level $e_2 \neq e_1$. Firms, seeing education level e_1 , know that the worker is of type 1, and pay a wage e_1 . And if they see education level e_2 , they offer a wage of $2e_2$, because they correctly assume that worker is of type 2.

In the second type of equilibrium, workers all pool at a single education level; this is called a *pooling equilibrium*. All the workers choose some education level e^* , at which point wages are $1.5e^*$.

The term *screening equilibrium* is often used interchangeably with separating equilibrium; cf. p. 478. In addition to these pooling and separating equilibria, there are other types, which you will explore in problem 2.

^a If we wanted to be fancy, we would permit general probability distributions, replacing (1) with a statement about e in the support of the distribution and turning (2) into the appropriate statement about conditional probabilities. But this generality would be wasted. If you start with a general formulation, you can prove in this setting, with strictly concave u_t that no type would ever select more than a finite number of education levels; see problem 2.

⁶ To be pedantic: $.5\pi_1(e)$ is the joint probability that a worker is of low ability and he chooses education level e obtained as the product of the marginal probability that a worker is of low ability, $.5$, and the conditional probability that a worker chooses education level e given he has low ability, $\pi_1(e)$. Hence $.5\pi_1(e) + .5\pi_2(e)$ is the marginal probability that a worker chooses education level e . And, by the rules of conditional probability, $.5\pi_1(e)/[.5\pi_1(e) + .5\pi_2(e)]$ is the conditional probability that a worker is of low ability given he chooses education level e .

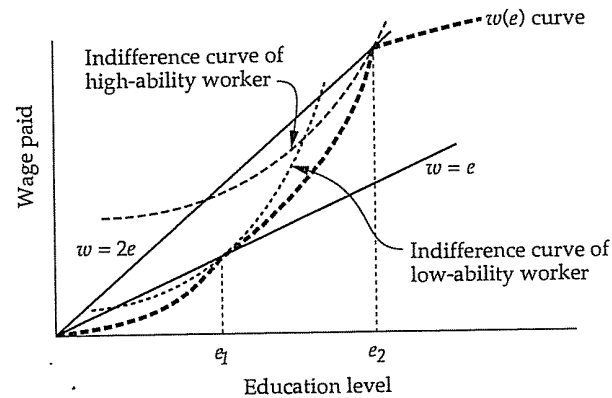


Figure 17.4. A separating equilibrium.

In this picture and in all others to follow, the heavy dashed curve indicates the workers' conjectures about what wages they will receive at every given level of education.

Return to the description of a separating equilibrium. For this to be an equilibrium, we need that workers, anticipating all this, are content to be separated in this fashion. That is, a worker of type 1 would rather choose the wage-education pair $(w = e_1, e = e_1)$ than $(w = 2e_2, e = e_2)$, and vice versa for workers of type 2. You'll see just such a situation in figure 17.4.

In this figure, note carefully the heavy dashed curve. This represents the function $w(e)$, workers' conjectures about the wages they will receive as a function of the level of education they select. Note that this function must lie everywhere at or below the indifference curves of the workers at points that the workers select; this is the self-selection condition. Condition (2) for an equilibrium pins down the $w(e)$ curve at the two levels of education selected, as shown.

This does not restrict the curve $w(e)$ at points e that are not selected in equilibrium. What restrictions might we wish to place on $w(e)$ at such points? For now, consider the following three possibilities:

(a) No restrictions whatsoever are placed on $w(e)$ at points e that are not selected in equilibrium. (We require throughout that $w(e)$ is nonnegative for every value of e ; that is, the institution of slavery is not part of our economy.)

17.2. Signaling quality

(b) We require that $e \leq w(e) \leq 2e$ for all e .

(c) We require (b) and, in addition, that $w(e)/e$ is nondecreasing.

What might motivate us to assume (b) or (c)? We reason as follows. A worker who chooses education level e is worth either e or $2e$ to a firm. The worker cannot possibly be worth more than $2e$ or less than e . Now if firms are unsure of the worker's level of ability, firms cannot be certain which of e or $2e$ is correct. And so, depending on their assessments for the chances that the worker is worth e or $2e$, they might be willing to bid anywhere between these two levels. But they would never pay more than $2e$, and the forces of competition between firms mean that they could never get away with paying less than e . Workers anticipate all this, so they anticipate that $e \leq w(e) \leq 2e$. As for (c), take this a step further and imagine that workers conjecture that firms will assess that a worker who chooses education level e will be of low ability with probability $\alpha(e)$. Assuming firms are risk neutral (which we do) and that they share assessments, competition amongst them would lead them to bid $w(e) = \alpha(e)e + (1 - \alpha(e))2e$. (Note that this is just another way of giving our justification for [b].) And now (c) can be translated as: $\alpha(e)$ is nonincreasing. Or firms' conjectures as to the ability of a worker do not give a higher probability for a lower-ability worker if the worker obtains more education.

At the risk of (re)stating the obvious, let us rephrase the equilibrium condition (2) in these terms. We require that firms' assessments given by $\alpha(e)$ are confirmed at education levels that workers do select in equilibrium. Or, in symbols,

$$\alpha(e) = \frac{.5\pi_1(e)}{.5\pi_1(e) + .5\pi_2(e)}.$$

If we imagine a stable population with new workers coming along and presenting themselves for employment each year, and if we imagine that the equilibrium distribution of workers' abilities and education levels is stable, then we would be saying that firms, based on their experience, know that distribution. What we do not restrict, or restrict only minimally, are the conjectures of firms at education level choices that are not chosen and, therefore, at which the firms have no experience.

You may be thinking that we shouldn't speak directly about the firms' conjectures concerning the ability of a worker who chooses a nonequilibrium level of education. The function $w(e)$ represents the workers' conjectures about the wages they will be paid. Hence, at least, we should speak of

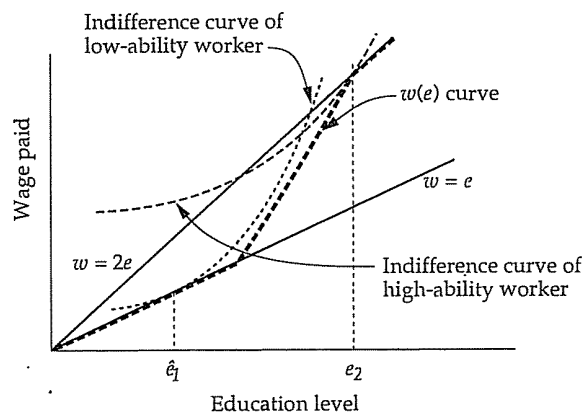


Figure 17.5. Another separating equilibrium.

the workers' conjectures concerning the firms' assessments about the ability of a worker who chooses a nonequilibrium level of education. And then, by having a single $w(e)$ function, we have implicitly assumed that all the workers have the same conjectures about the firms' assessments, and this conjecture holds that all the firms will have the same assessments. Are these assumptions of coincident conjectures and assessments necessary? If you are very careful, you should be able to see that much of what we will say does not depend on this, although we will hold to this implicit assumption to keep the exposition simple. You may also wonder why we have not introduced game-theoretic terminology for all of this. We will do so in the next section, but there are reasons to put this off for a while.

Note that the function $w(e)$ in figure 17.4 does not satisfy (b) or (c), although it does satisfy one somewhat obvious condition, namely that it is nondecreasing. In figure 17.5 we give a second (separating equilibrium) in which $w(e)$ does satisfy (b).

Note in figure 17.5 the point chosen by the low-ability workers, denoted by $\hat{e}_1$. This, as drawn, is the point along the ray $w = e$ that low ability workers like most; notice that their indifference curve through this point is tangent to $w = e$. In contrast, in figure 17.4, low-ability workers get less utility than they would get from $(w = \hat{e}_1, e = \hat{e}_1)$. We use $\hat{e}_1$

throughout this chapter to denote this particular point of tangency, and we have the following result:

Proposition 1. In any separating equilibrium for which the function $w(e)$ satisfies (b), low-ability workers choose precisely $\hat{e}_1$ and get the corresponding wages. In any equilibrium (separating or not) for which the function $w(e)$ satisfies (b), low-ability workers get at least the utility that they get from $(w = \hat{e}_1, e = \hat{e}_1)$.

You should have no problems at all seeing why this is true. Or, rather, you should have no problems with the first part of this proposition. The second part may be a tiny bit harder.

Pictures of pooling equilibria are relatively easy to draw. In a pooling equilibrium, recall, we assume that all the workers choose a single education level e^* . As before, firms' beliefs, seeing e^* , must be confirmed. Since we have supposed that the numbers of high- and low-ability workers in the population are equal, and since (in this equilibrium) all workers choose e^* , we see that in terms of the notation above $\alpha(e^*) = .5$, and $w(e^*) = 1.5e^*$. An equilibrium of this sort is given in figure 17.6. Note that the function $w(e)$ in this case satisfies (c). Note also that the function $w(e)$ has a kink at the pooling point. This is necessary; without a kink, we couldn't have $w(e)$ underneath both types' indifference curves and touching those curves at the pooling point.

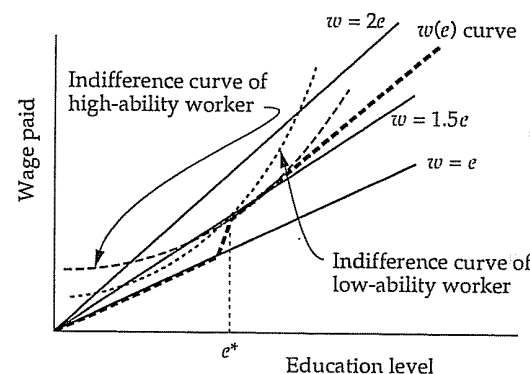


Figure 17.6. A pooling equilibrium.

As you can see (and imagine), there are a lot of equilibria here. Is there any reason to suspect that one is more plausible than another? We will get back to this question, but before doing so we look at another way to tell the basic story of market signaling.

The story of Rothschild and Stiglitz:

Worker self-selection from a menu of contracts

In the story just told, workers choose education levels in anticipation of offers from the firm, and (we assume) those anticipations are correct, at least for the education levels actually selected. Let's turn this around and suppose that the firms move first. Specifically, suppose that firms offer to workers a number of "contracts" of the form (w, e) before workers go off to school. Workers consider the menu of contracts on offer, sign the one they like best, and then go off to school, content in the knowledge of what wage they will get once school is done (assuming they complete the number of years of schooling for which they have contracted). This story may seem a bit lame in the context of education level choices. If you find it so, think instead of insurance markets, the context with which Rothschild and Stiglitz deal explicitly. Insurance companies are willing to insure one's life in any of a number of ways, with different premiums, death benefits, exclusions, and so on. A customer (shops for the policy) best suited to himself, given both his preferences and what is important here, the knowledge he has about his prospects for a long and healthy life.

An equilibrium in the sense of Rothschild and Stiglitz consists of (a) a menu of contracts, a set of pairs $\{(w^1, e^1), (w^2, e^2), \dots, (w^k, e^k)\}$ for some finite integer k ,^b and (b) a selection rule by which workers are "assigned" to contracts or, for each type t , a probability distribution π_t over $\{1, 2, \dots, k\}$, that satisfy three conditions:

(1) Each type of worker is assigned only to contracts that are best for that worker among all the contracts in the menu. In symbols, $\pi_t(j) > 0$ only if $u_t(w^j, e^j)$ achieves the maximum of $u_t(w^{j'}, e^{j'})$ for $j' = 1, \dots, k$.

(2) Each contract in the menu to which workers are assigned at least breaks even on average. (Otherwise, firms offering that contract would withdraw the contract.) In symbols, for each $j = 1, \dots, k$, if $\pi_1(j) + \pi_2(j) > 0$, then

$$w^j \leq \frac{.5\pi_1(j)}{.5\pi_1(j) + .5\pi_2(j)} e^j + \frac{.5\pi_2(j)}{.5\pi_1(j) + .5\pi_2(j)} 2e^j.$$

^b We assume that the menu of contracts offered is finite mostly for expositional convenience.

(3) No contract can be created that if offered in addition to those in the menu would make strictly positive profits for the firm offering it, assuming that workers choose among contracts in a manner consistent with rule (1) above.

We won't try to write (3) out formally — it makes a good exercise if you are fascinated by symbol manipulation.^c

This change in formulation has dramatic effects. To begin the analysis, we assert something common to the Spence formulation:

Proposition 2. *In an equilibrium, any contract that is taken up by workers must earn precisely zero expected profits per worker.*

The proof of this is a little tricky.⁷ A natural argument to try runs as follows: Suppose that (w', e') is offered, is taken by some workers, and earns an expected profit of size ϵ per worker who took it. Then have some firm offer $(w' + \epsilon/2, e')$. This new contract will attract all the workers who previously were attracted by (w', e') and will still return a profit of $\epsilon/2$, contradicting condition (3) of an equilibrium. The trouble with this argument is that the new contract may attract others besides those who previously took (w', e') , and those others may render this contract unprofitable.

Only a sketch of the correct proof will be given, with details left for the reader to fill in. First, the argument just given works if the contract (w', e') attracts only low-ability workers. Then if $(w' + \epsilon/2, e')$ manages to attract any high-ability workers, it is even more profitable. So the hard case is where (w', e') attracts some high-ability workers. At this point, wait to read through the proof of proposition 3, which shows how to break a pooling equilibrium. The key to that argument is that it is possible to devise a contract $(w' + \epsilon/2, e' + \delta)$, for $\delta > 0$ that is more attractive to high-ability workers than is (w', e') but is less attractive to low-ability workers. Hence if a firm offers this contract, it will attract all the high-ability workers who previously chose (w', e') , leaving the low-ability workers at (w', e') (or somewhere else). If (w', e') made profits ϵ per worker, then this new contract makes profits exceeding $\epsilon/2$ per worker.

^c And if you do try to write this out formally, you will run up against the following question: Should we insist that the contract earn nonpositive profits for every assignment consistent with (1), or just that it make nonpositive profits for some assignment consistent with (1)? It won't matter to the theory which you choose, as long as you are careful about the arguments that follow.

⁷ In Spence's formulation, it is true by fiat.

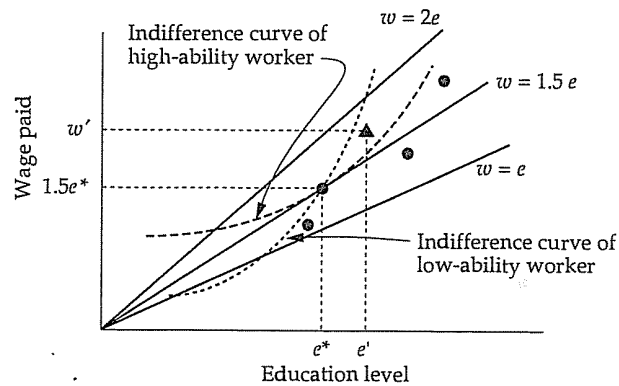


Figure 17.7. Destroying a pooling equilibrium.

Now we give a result that distinguishes this model from that of Spence:

Proposition 3. *It is impossible that an equilibrium is a pooling equilibrium.*

The argument runs as follows: Consider figure 17.7, and the pooling equilibrium depicted. Suppose the firms are offering a menu of contracts that causes all the workers to choose $(1.5e^*, e^*)$ as shown. In this figure, the filled-in dots are the menu of contracts that we suppose are offered. We have added a few contracts to the menu besides $(1.5e^*, e^*)$, although they are irrelevant, since (in our pooling equilibrium) the contract $(1.5e^*, e^*)$ is best for all types of worker.

From this position, any one of the firms can offer the contract (w', e') that is shown as a filled-in triangle. Saying in words what the picture shows: This is a contract that has slightly higher wages and education levels than $(1.5e^*, e^*)$, where the increased wages more than compensate a high-ability worker but don't compensate a low-ability worker relative to the pooling contract. Since this contract is added to the contracts already in the menu, it will only attract the high-ability workers; all the low-ability workers prefer the e^* offer. But if all the high-ability workers flock to (w', e') , and none of the low-ability workers do so, each worker who chooses (w', e') will be worth $2e'$ to the firm that makes this offer. The firm makes a profit, and we can't have an equilibrium.

The same sort of argument can be used to establish

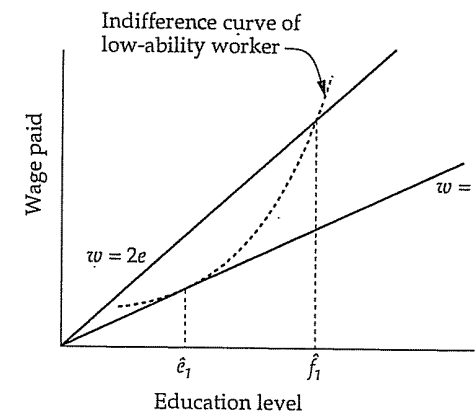
Proposition 4. *It is impossible, in equilibrium, that any contract (w, e) is taken by positive fractions of the high- and low-ability workers both. Or, in other words, the only possible equilibria are separating equilibria.*

The details are left to you; a picture very much like figure 17.7 is the key. Next we establish

Proposition 5. *There is a single candidate for a separating equilibrium. In this candidate equilibrium, low-ability workers choose the contract $(\hat{e}_1, \hat{e}_1)$, where $\hat{e}_1$ is defined as before, and ... (to be continued)*

Why must this be so? Because if low-ability workers are separated and are choosing any other contract, a firm could add to the menu of contracts a contract $(w = \hat{e}_1 - \delta, e = \hat{e}_1)$ for $\delta > 0$ small enough so that low-ability workers prefer this to the contract they are choosing in the supposed equilibrium. But then for any $\delta > 0$, this contract must be profitable.

Fixing $\hat{e}_1$ at this value, define the education level $\hat{f}_1$ to be that level of education such that low-ability workers are indifferent between $(w = \hat{e}_1, e = \hat{e}_1)$ and $(w = 2\hat{f}_1, e = \hat{f}_1)$. Figure 17.8 shows this point. And let $\hat{f}_2$ solve the problem: Maximize $u_2(2e, e)$ subject to $e \geq \hat{f}_1$. That is, $\hat{f}_2$ is the

Figure 17.8. Determination of $\hat{f}_1$.

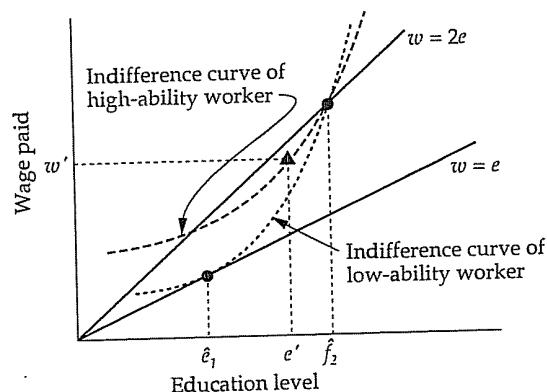


Figure 17.9. A separating equilibrium.

education level that high-ability workers would choose if they could have any point along the ray $w = 2e$ for $e \geq \hat{f}_1$.

Proposition 5, continued. ... and (in the single candidate separating equilibrium) high-ability workers get $(w = 2\hat{f}_2, e = \hat{f}_2)$.

Why? Because if they are separated at any other contract, some firm could come along and offer a contract $(w = 2\hat{f}_2 - \delta, e = \hat{f}_2)$ that for δ sufficiently small would be more attractive to high-ability workers than the contract they are taking at the supposed equilibrium. Since $\hat{f}_2 \geq \hat{f}_1$, this contract is less appealing than $(w = \hat{e}_1, e = \hat{e}_1)$ for the low-ability workers. Hence this contract will attract precisely the high-ability workers. And thus for any $\delta > 0$, this contract is strictly profitable.

Consider figure 17.9. We have there depicted the proposed separating equilibrium in a case where $\hat{f}_2 = \hat{f}_1$. It may be helpful to say why no firm would try to break this equilibrium by offering a contract in a position such as (w', e') that is shown, with w' a bit less than $2e'$ and e' a bit less than $\hat{f}_1$. This contract, added to the menu, would certainly attract high-ability workers. And every high-ability worker attracted would be profitable at this wage. But it would also attract all the low-ability workers in proportion to the population, if you attract high- and low-ability workers in proportion to the population, you have to pay less than 1.5 of the education level they pick to make a profit.

Now consider figure 17.10. This is just like figure 17.9, except that the indifference curve of the high-ability workers through the point $(w = 2\hat{f}_2, e = \hat{f}_2)$ dips below the line $w = 1.5e$. In this case, a firm could offer a contract such as (w', e') as shown, below the line $w = 1.5e$, but still above the high-ability worker's indifference curve. This would break the equilibrium, because even though it attracts all the workers, high-ability and low-, it is still profitable.

In figure 17.10, then, there can be no equilibrium at all. Any sort of pooling is inconsistent with equilibrium. And the only possible separating equilibrium can also be broken. In contrast, although we won't go through all the details, in situations such as figure 17.9 (or situations where $\hat{f}_2 > \hat{f}_1$), the candidate separating equilibrium is an equilibrium. Summarizing all this:

Proposition 6. In the formulation of Rothschild and Stiglitz, there is at most one equilibrium. In the candidate equilibrium, low-ability workers choose $(w = \hat{e}_1, e = \hat{e}_1)$ and high-ability workers choose the education level $\hat{f}_2$ such that $(w = 2\hat{f}_2, e = \hat{f}_2)$ is their most preferred point along the ray $w = 2e$ for $e \geq \hat{f}_1$. If the indifference curve for high-ability workers through the point $(w = 2\hat{f}_2, e = \hat{f}_2)$ dips below the pooling line $w = 1.5e$, then there is no equilibrium at all. If this indifference curve stays above (or just touches) the pooling line, then this single candidate equilibrium is an equilibrium.

Quite a difference from Spence's model, where there were many possible equilibria!

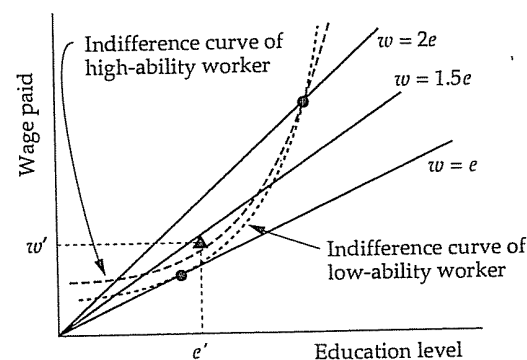


Figure 17.10. No equilibrium at all.

The stories of Riley and Wilson

In many markets where signaling takes place, such as insurance, it seems natural to suppose that the uninformed parties put a menu of offers "on the table" from which informed parties must choose. So the conclusion of the Rothschild and Stiglitz analysis, namely that there may be no equilibrium at all, is rather troubling.^d A number of authors have suggested that the "flaw" in the analysis lies in the notion of equilibrium proposed by Rothschild and Stiglitz. In particular, it is implicit in the equilibrium notion that if a new contract is added to the menu, nothing further (concerning the menu) changes. In place of this, one might imagine that firms already offering contracts would react in one way or another, possibly rendering previously profitable deviations unprofitable and restoring particular equilibria.

Two of the original treatments of this are due to Riley (1979) and Wilson (1977). Riley advances the notion of a *reactive* equilibrium, in which, to destroy a proposed equilibrium, it must be possible to add a contract to the menu that will be strictly profitable and that will not become strictly unprofitable if other firms are allowed to add still more contracts to the menu.^e In this sort of scheme, our argument for why there are no pooling equilibria stands up; we broke a pooling equilibrium by adding a contract that attracted high-ability workers only, given that the pooling contract remained to attract low-ability workers. There is no way to make this pool-breaking contract strictly unprofitable as long as the old pooling contract remains, because as long as that old pooling contract is present, only high-ability workers would ever consider taking the new contract. But you are much more at risk when adding a contract that attempts to break a separating equilibrium by pooling high- and low-ability workers. For in this case, still another contract can be added that siphons from your contract the high-ability workers, leaving you (the original defector) with only low-ability workers. Hence, Riley concludes, reactive equilibria always exist, and they correspond to the single candidate separating equilibrium.

Wilson, on the other hand, proposes a notion called an *anticipatory* equilibrium. In this case, to break a proposed equilibrium it must be possible to add a contract that is strictly profitable and that doesn't become strictly unprofitable when (now) unprofitable contracts from the original menu are *withdrawn*. Now if you try to break a pooling equilibrium by

^d It becomes more troubling if you consider a population of workers whose education levels are continuously distributed. Then nonexistence of equilibrium is the "usual" state of affairs; cf. Riley (1979).

^e The discussion given here is a bit rough. Consult the literature to get more precise statements of these results.

skimming the high-ability workers, you are at risk; the pooling contract that you destroy becomes unprofitable because after your addition it attracts only low-ability workers. But if it is withdrawn, all these low-ability workers may flock to your contract. On the other hand, if you break a separating equilibrium with a pooling contract, you are not at risk; because you have created a contract that attracts all the workers, you don't care if other, now unused contracts are withdrawn. So, Wilson concludes, *anticipatory equilibria* always exist; sometimes there is more than one; and pooling is possible as an equilibrium outcome.

How do we sort between Rothschild and Stiglitz, Riley, and Wilson? This clearly depends on what you think are reasonable assumptions in a given market concerning firms' abilities to add or withdraw contracts from what is offered to the public. Think in terms of an insurance market that is regulated by some regulatory authority. If you think that firms cannot withdraw contracts because (say) the regulatory authority forbids this, then Riley's equilibrium concept seems the more reasonable. If you think that the regulatory authorities permit the firms to withdraw unprofitable contracts and are very tough on adding contracts that potentially affect the profitability of others, then Wilson's notion seems quite reasonable. As for Rothschild and Stiglitz, think of regulators who call for firms to register simultaneously and independently the contracts they wish to offer, with no room to add or subtract subsequently. (Do any of these sound particularly realistic to you? If not, see the discussion in section 17.4.)

17.3. Signaling and game theory

If you paid attention to part III of this book, you should have no difficulty guessing where the preceding paragraph is taking us. We can try to sort through the different equilibrium concepts by sorting through details of the market institutions. Perhaps by specifying precise "rules of the game" we can see when one equilibrium notion or another is more relevant.

Spence's model in game theoretic terms

It is easy to distinguish between Spence on the one hand and Rothschild and Stiglitz on the other. For Spence's story, consider the following (extensive form) game with incomplete information. A worker's abilities are determined by nature and are revealed to the worker. That is, this is a game of incomplete information about the worker's level of ability. The worker, based on this, determines how much education to obtain. The level of education obtained, but not the worker's abilities, is observed by

a number of firms, who then bid in Bertrand fashion for the worker's services.^e

The Nash equilibria of this game are precisely the equilibria of Spence if we do not add any restrictions to $w(e)$ beyond the basic (equilibrium) restriction that $w(e)$ must be "correct" for levels of education that are chosen in equilibrium by the worker. Firms' responses to other choices of e are out-of-equilibrium responses, since the worker wasn't supposed to pick those levels of e , and (as you will recall) the Nash equilibrium criterion doesn't restrict out-of-equilibrium responses. We restrict those responses by refining the Nash concept. Subgame perfection isn't much use in this case; this game of incomplete information has no proper subgames. But the notion of sequential equilibrium is quite helpful. For example, if we require that the equilibrium is sequential, then we get precisely restriction (b) on the $w(e)$ function; at every level of education, including those that are out-of-equilibrium, each firm must have out-of-equilibrium beliefs $\alpha(e)$ that the worker is low ability, and must bid accordingly.^f And restrictions such as (c) that $\alpha(e)/e$ must be nonincreasing are straightforward restrictions on beliefs; restriction (c), for example, says that the probability assessed by a firm that a worker is of high ability is nondecreasing in the level of education the worker chooses.

In fact, high-powered restrictions on out-of-equilibrium beliefs are very powerful in this sort of application. Let us pose here three increasingly stiffer restrictions, and then see where they lead:

(1) *The equilibrium should be sequential.*

(2) *Suppose a type t of worker and education levels e and e' are such that $u_t(e, e) > u_t(2e', e')$. Then in any Nash equilibrium, it must be possible to sustain the equilibrium outcome with beliefs that put probability zero on education level e' being selected by type t .*

(3) *Fix a sequential equilibrium, and let u_t^* be the equilibrium level of utility achieved in this equilibrium by a worker of type t . Suppose that for some type t and education level e' , $u_t(2e', e') < u_t^*$. Then it must be possible to sustain the given equilibrium outcome with beliefs that put probability zero on education level e' being selected by type t .*

^e By having the firms bid in Bertrand fashion, we are guaranteed that in any equilibrium the worker will be paid his conditional expected value, conditional on the level of education he selects. This is true in any Nash equilibrium for education levels that are selected with positive probability in the equilibrium, and it is true in any sequential equilibrium for any education level at all, where we compute conditional expected values using the firms' beliefs.

^f The fact that the firms all have the same out-of-equilibrium beliefs derives from the consistency requirement of a sequential equilibrium. If we dropped this for something weaker, the "answers" would not change much, but the analysis would be a bit trickier.

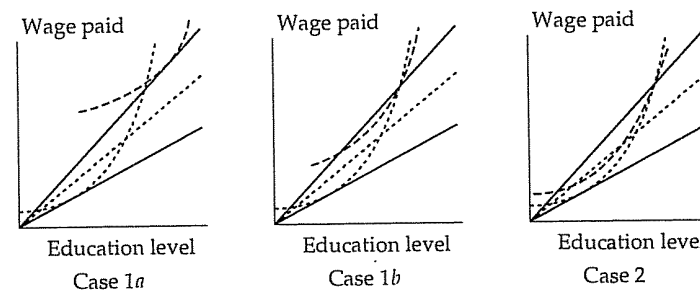


Figure 17.11. Three cases for market signaling.

Requirement (1) is entirely straightforward. The motivation behind (2) runs as follows: Fix t , e , and e' as postulated. In any sequential equilibrium a type t worker is guaranteed utility level at least $u_t(e, e)$ if he picks education level e . If he picks e' , the best he could conceivably do is $u_t(2e', e')$. So if he anticipates sequentially rational responses by employers at all levels of education, he has no business whatsoever in picking education level e' ; he is guaranteed a better outcome if he picks e . Employers should be able to figure this out and, therefore, assess probability zero that the choice of education level e' comes from t . Requirement (2) is that the equilibrium can be sustained with beliefs so restricted.

Requirement (3) takes (2) another step. There the test is, Is the best that type t can do if he chooses e' worse than what he gets in equilibrium? If so, requirement (3) is that employers should be able to assess probability zero that e' comes from t .

Note well: Requirement (2) is something like elimination of dominated strategies, where we require that a strategy be dominated by another only at sequentially rational responses by firms. Requirement (3) is stronger; it is of the category of elimination of strategies that are *equilibrium dominated* (cf. subsection 12.7.4). Taking the extra step from (2) to (3) is controversial; see the brief discussion and references given in chapter 12.

What are the consequences of these requirements? Recall from our discussion above that $\hat{e}_1$ is the level of education that maximizes u_1 along the ray $w = e$; $\hat{f}_1$ is the level of education such that $u_1(\hat{e}_1, \hat{e}_1) = u_1(2\hat{f}_1, \hat{f}_1)$; and $\hat{f}_2$ is the education level most preferred by the type two workers along the ray $w = 2e$ for $e > \hat{f}_1$. (See figure 17.8.) We distinguish between two cases: In case 1 the high-ability worker's indifference curve through $(w = 2\hat{f}_2, e = \hat{f}_2)$ stays above the pooling line $w = 1.5e$; and in case 2 the high-ability worker's indifference curve through this point passes below the pooling line. Case 1 can be divided into two subcases: Case 1a, where $\hat{f}_2 > \hat{f}_1$; and case 1b, where $\hat{f}_2 = \hat{f}_1$. In figure 17.11, we depict all three cases for purposes of comparison.

Proposition 7. *For market signaling formulated as a game in which workers choose education levels and then firms bid in Bertrand-like fashion:*

(a) If we insist on our equilibrium being sequential (requirement [1]), then in any separating equilibrium the low-ability workers select education level $\hat{e}_1$. More generally, in any sequential equilibrium, low-ability workers obtain utility at least equal to $u_1(\hat{e}_1, \hat{e}_1)$.

(b) If we insist in addition that requirement (2) is met, then the only possible separating equilibrium has high-ability workers choosing education level $\hat{f}_2$. In general, in any sequential equilibrium that meets requirement (2), high-ability workers must obtain utility at least equal to $u_2(2\hat{f}_2, \hat{f}_2)$. And in case 1, the separating equilibrium is the only possible sequential equilibrium that meets requirement (2), separating or not.

(c) The only sequential equilibrium that meets requirement (3) is the separating equilibrium in which low-ability workers choose $\hat{e}_1$ and high-ability workers choose $\hat{f}_2$.

Let us sketch the proof. If the equilibrium is sequential, then wages in response to any education level e cannot be less than e or more than $2e$. Hence if low-ability workers choose $\hat{e}_1$, they are assured of utility $u_1(\hat{e}_1, \hat{e}_1)$. This implies (a) immediately.

For part (b), consider any education level $e' > \hat{f}_1$. By the definition of $\hat{f}_1$, $u_1(\hat{e}_1, \hat{e}_1) > u_2(2e', e')$. Hence if requirement (2) is imposed, we must be able to sustain the equilibrium with beliefs that education levels $e' > \hat{f}_1$ are certainly from high-ability workers, which means they command wages $2e'$. Hence for any $e' > \hat{e}_1$, a high-ability worker is sure to get utility $u_2(2e', e')$. In any equilibrium, a high-ability worker need never accept utility less than $\sup\{u_2(2e', e') : e' > \hat{f}_1\}$, which implies that he must get utility at least equal to $u_2(2\hat{f}_2, \hat{f}_2)$. This is the second part of (b). (You may need to think a bit about this step.) In any separating equilibrium, a high-ability worker can't separate at a level of education less than $\hat{f}_1$ (Why not?), and so we have the first part of (b).

The third part of (b) is the hardest and is left to the virtuous reader with the following hint: We know that at any equilibrium outcome, low-ability workers sit on an indifference curve on or above their curve through $(\hat{e}_1, \hat{e}_1)$. Consider separately the case where they sit along the indifference curve through $(\hat{e}_1, \hat{e}_1)$ (in which case apply the second part of [b]) and the case where they sit above this curve, in which case (we assert and you must show) they must all be part of a single pool, which leads to a contradiction.

As for (c), the key is to show that any pooling at all in equilibrium will not meet requirement (3). Consider figure 17.12 and the point shown with the dot. Imagine that both high- and low-ability workers pooled at that point. Then, in the equilibrium, low-ability workers must obtain the utility level that parameterizes the indifference curve through that point. Consider education levels between the points $\hat{g}_1$ and $\hat{g}_2$ shown in the figure. At any education level $g > \hat{g}_1$, the low-ability worker cannot do better than he does at his equilibrium level of utility, even if wages $2g$ are paid. So by requirement (3), we must be able to sustain this equilibrium with beliefs that attribute no chance to education levels $g > \hat{g}_1$ coming from low-ability workers. Hence wages $2g$ will be paid for such education levels. And then a high-ability

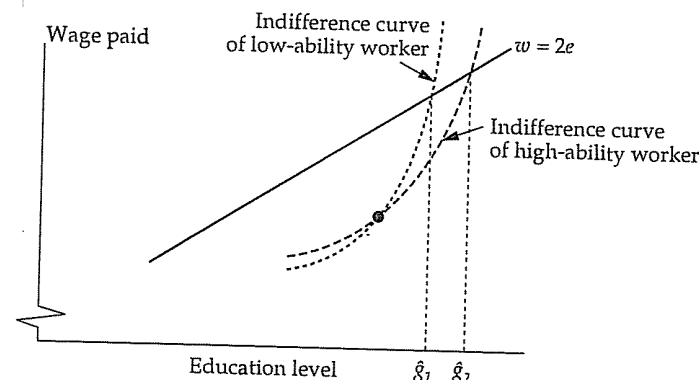


Figure 17.12. Breaking a pooling equilibrium with requirement (3).

worker would rather choose some such g that is less than $\hat{g}_2$ than stick to the supposed pool. Done.

Rothschild and Stiglitz and game theory

Rothschild and Stiglitz, on the other hand, can be understood as an analysis of the following alternative extensive form game. Firms simultaneously and independently name contracts (in wage-education space, in our example) that they are willing to honor. Then the worker responds by taking one of the contracts (or refusing them all).⁹ The equilibria of Rothschild and Stiglitz correspond to pure strategy, subgame perfect Nash equilibria of this game. In particular, when there are no Rothschild and Stiglitz equilibria, this game has no pure strategy, subgame perfect Nash equilibria.⁹

Further variations

For Riley and for Wilson, we might imagine game forms of the following sorts. For Riley, firms simultaneously and independently propose

⁹ By invoking subgame perfection, we can and hereafter do assume that the worker chooses the contract that he likes best from those that are offered. We can use subgame perfection here because Nature's move can be put in after the firms offer their contracts, and at that point Nature's move starts a proper subgame.

⁹ The game always admits Nash equilibria, in mixed strategies if there are no pure strategy equilibria. (The existence result behind this claim is not trivial.) Whether one believes that the mixed strategy equilibria have any particular relevance is another matter.

contracts. The set of contracts offered are made public. Then firms are given the opportunity to add contracts. If anyone adds a contract, firms are given a further opportunity to add, and so on, until no firm adds any contracts. And then workers choose their preferred contracts. (What if firms add contracts perpetually? I will leave this question hanging for now.) For Wilson, imagine that firms simultaneously and independently propose contracts. These contracts are made public. And then firms are allowed to remove contracts from the offer set. After one or more rounds of removals, workers choose.

The game form proposed for Riley is incompletely specified, because we haven't said how the game ends if firms keep adding contracts. And, in any case, this game form admits too many Nash equilibria. Consider, for example, an equilibrium where firms initially offer to pay workers much less than they are worth. We would imagine a single firm breaking this equilibrium by offering slightly better terms and getting all the workers. But now firms can respond, and they might be supposed to respond in a way that leaves all the firms making zero profits. In which case no firm would have a positive incentive to deviate from the first, positive profits situation. In a sense, we have "implicit collusion" among the firms, supported by supergame-like threats of reaction. (On this point and others concerning this game form and its relation to Riley's reactive equilibrium, see Engers and Fernandez [1987].)

We leave it to you to connect the game form advanced for Wilson's equilibrium notion to that notion; you will (clearly) have to study Wilson's paper first.

To close this discussion, we mention yet another game form that with high-powered refinements of Nash equilibrium leads to a strong result. Imagine a three-stage game constructed as follows. Firms make offers, just as in Rothschild and Stiglitz. Workers choose their education levels and queue for jobs at the firms whose offers they wish to take. But then firms can renege; having seen which other contracts were offered and which choices were made, firms can decide not to honor offers they made initially. Hellwig (1986) studies this game form and obtains results that can be paraphrased roughly as: If one looks for *stable equilibrium* outcomes in the sense of Kohlberg and Mertens (1986), then Wilson-style equilibria are selected.

17.4. Bibliographic notes and discussion

Several messages can be obtained from the preceding analyses. First, "lemons problems" can exist in markets where one party to a transaction has available information that a second party lacks.

Second, these problems may be ameliorated through market signaling or self-selection by the side having the private information. If the informed side takes the active role as in Spence's analysis, this is known as *market signaling*. If the side without information proposes a menu of contracts among which the informed side selects, this is conventionally called *market screening*. The "instrument" used to send this signal should have the property that it is (relatively) cheaper for "higher quality" parties to convey; i.e., something analogous to the single-crossing property should hold. (Higher quality parties are "usually" adversely affected by the presence of lower quality parties; either the higher quality parties are pooled in with lower quality parties, to their detriment, or they must invest in signals beyond the point that they would if there were no informational asymmetry to distinguish themselves from their lower quality peers. (I put "usually" in quotes here because this result must be made precise. It is a substantial challenge, but you might wish to try your hand at turning this vague assertion into a precise result.)

The range of applications for this sort of signaling or screening is quite large. Besides education as a signal (Spence, 1974) and signaling in insurance (Rothschild and Stiglitz, 1976), a partial list of applications is: a manufacturer offers warranties as a signal that the goods in question are of high quality (Grossman, 1981); a plaintiff demands a high pretrial settlement as a signal that her case is particularly strong (Reinganum and Wilde, 1986; Sobel, forthcoming); a bargainer, to signal relative bargaining strength, rejects the other side's offer (Rubinstein, 1985) or delays making his own offer (Admati and Perry, 1987); and a monopolist, to signal low costs and so to forestall entry by potential rivals, charges a low price before entry occurs (Milgrom and Roberts, 1982). You may recall this last application from chapter 13. Lest you think you've seen the whole story, there is much more to entry deterrence as signaling than the simple analysis in chapter 13 conveys. In particular, Milgrom and Roberts show how the incumbent monopolist could engage in entry deterring pricing that is completely ineffective in deterring entry. You may also recognize the beer-quiche example of chapter 13 as another example of signaling.

The relationships between the early models of market signaling and game theory have been developed only recently. Stiglitz and Weiss (forthcoming) should be consulted concerning the basic differences between signaling and screening models. Work on signaling models (i.e., on Spence-style models) stressing restrictions on out-of-equilibrium beliefs can be found in Banks and Sobel (1987), Cho and Kreps (1987), and Cho and Sobel (1988) among others. Less work has been done on connecting screening

models to game theory, although Engers and Fernandez (1987) and Hellwig (1986) can be consulted for some initial work.

The topic of market signaling fans out in many directions from the very simple and basic model that has been discussed here. In the first place, one might think of looking at models with more than two types of employee. This sort of extension is relatively straightforward, and most of the references given above do this. More interesting are applications where more than one signal is available (see Engers [1987] for screening models; and see Cho and Sobel [1988] for signaling models), or where the "quality" being signaled is multidimensional (Quinzii and Rochet, 1985), or where signals are noisy so that the act of signaling by itself is a superior signal (Hillas, 1987).

An important variation that we have not touched upon concerns signaling models where the signals are costless to send. That is, participants can engage in "cheap talk" — things costless to say, but not at all inconsequential to the equilibria that can be attained. See Crawford and Sobel (1982) and Farrell (1988) on this topic. And throughout this chapter we have been concerned with the signaling of private information. One can also analyze the signaling of previously taken actions, or even the signaling of actions intended in the future. (See, for example, Wolinsky [1983] or Ben-Porath and Dekel [1989].)

An issue less theoretical and more "practical" concerns the impact of repeated play in the oligopolistic sense on equilibria in the context of screening. That is, if we imagine insurance companies that offer potential clients a menu of contracts, and we think that these companies might engage in some form of implicit collusion, then the discussion of Rothschild and Stiglitz versus Riley versus Wilson might be turned on its head. Perhaps the companies collude and offer a menu of contracts that is best for them as a cartel, held together by a threat of reversion to cut-throat competition. This leads one to ask, If the insurance business is not competitive but instead is a monopoly (or a cartel), what menu of contracts will be offered? (And if the insurance business is regulated, what then?) With this as a lead-in we move on to chapter 18.

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17.5. Problems

- 1. Recall the lemons problem at the start of the chapter. Some used cars are peaches, worth \$3,000 to buyers and \$2,500 to sellers, and some are lemons, worth \$2,000 to buyers and \$1,000 to sellers. There is a fixed supply of cars and unlimited demand. Suppose there are twice as many peaches as lemons. Assume buyers can't tell the quality of a given car, while sellers can. What do the supply and demand curves look like? In the text, we asserted that markets clear at \$2,666.67. Are there other market clearing prices?
- 2. Consider the model of job market signaling explored in section 17.2. More specifically, consider the mode of analysis that is attributed to Spence, where workers choose education levels in anticipation of the wages they will receive. We explored in section 17.2 both separating and pooling equilibria, where we assumed that each type of worker chooses one and only one level of education in equilibrium.
 - (a) Are there any separating equilibrium where one type (or both) chooses more than one level of education? By a separating equilibrium we mean that if one type chooses a given education level with positive probability, then the other type does not.
 - (b) Is there any pooling equilibrium where both types choose more than one level of education? By a pooling equilibrium we mean that every education level chosen by one type with positive probability is chosen by the other type with positive probability.
 - (c) A hybrid equilibrium is one in which some education levels are chosen by one type only and others are chosen by both types. Are there any hybrid equilibria? If so, how many different "types" of them can you construct?
 - (d) What is the maximum number of education levels that a low-ability worker can choose with positive probability in any equilibrium? What is

the maximum number of education levels that a high-ability worker can choose with positive probability in any equilibrium? Justify your answers.

- 3. (a) Complete the proof of proposition 2.
- (b) Prove proposition 4.
- 4. (a) Consider the model of job market signaling explored in section 17.2, but with one modification: Education is completely worthless in terms of increasing worker productivity. That is, a low-ability worker is worth v_1 to a firm and a high-ability worker is worth v_2 , regardless of how much education either has. We assume that $v_1 < v_2$. Otherwise, things are as in section 17.2; workers like higher wages and dislike education, their indifference curves have the "single-crossing property," and so on. Recount the conclusions of Spence and then of Rothschild and Stiglitz in this modified formulation.
 - (b) In the modified formulation, it is clear that any education obtained is socially wasteful; it doesn't improve productivity and it lessens the utility of the worker. Put another way, if workers obtain any education at all in equilibrium, they are clearly being overeducated from the point of view of enhanced productivity by society. Presumably you showed in part (a), therefore, that equilibria in this setting can result in "too much" education, and separating equilibria must do so. In the original setting of section 17.2 do similar conclusions apply? Is it the case that there is never, in equilibrium, "too little" education? In other words, comment on the following assertion: "The model of education as a signal due to Spence and to Rothschild and Stiglitz shows that people overinvest in education, when education is used to signal ability."
- 5. Recall that we gave a very imprecise formulation of the single-crossing property. Suppose that the utility functions $u_t(w, e)$ are concave, monotonic in the right directions, and have derivatives of all orders. Can you give a more precise formalization of the single-crossing property that uses appropriate partial derivatives of the u_t ?
- 6. A quite abstract formulation of the single-crossing property can be given for the following setting: The worker is one of a finite number of types, indexed by $t = 1, 2, \dots, T$. The worker can choose education levels e from some interval, say $[0, \infty)$, and firms can pay wages w drawn from some interval, say $[0, \infty)$. We let $u_t(w, e)$ denote the utility obtained by a worker of type t from wages w and education level e . Then if, for some type of worker $t = 1, 2, \dots, T$, $u_t(w, e) \geq u_t(w', e')$ for $e > e'$, then

for all $t' = t + 1, t + 2, \dots, T$, $u_{t'}(w, e) > u_{t'}(w', e')$. In words, if a type t prefers (weakly) the package (w, e) to (w', e') , and the preferred package has higher education, then all types t' with higher index strictly prefer the package (w, e) .

Suppose we have a formulation of job market signaling as in Spence's story with finitely many types of workers as in the preceding paragraph. Suppose that we find an equilibrium. Assume that in this equilibrium workers conjecture that for each education level they might select, the wage that they will receive is nonrandom. (You saw no examples with random wages, so this last line may confuse you. If so, ignore it.) Assume the formulation of single-crossing just given and prove the following: If a type t worker obtains education level e with positive probability in the equilibrium, and a type t' obtains e' , then $t' > t$ implies $e' \geq e$.

■ 7. Consider the following game form representation of worker selection from a menu of contracts. The number of firms is finite. (Two will be enough.) Firms simultaneously and independently name contracts (pairs (w, e)); with each firm limited to a finite number of contracts. (If you wish, assume each firm can name no more than four contracts.) The contracts tentatively offered are announced, and then firms simultaneously and independently decide whether they wish to withdraw some of the contracts they offered initially. The new set of tentatively offered contracts is announced, and if any contracts were removed, firms are again given the opportunity to remove contracts. And so on, until either no contracts are left or, in a given round, no contracts are removed. (Since there are finitely many firms and finitely many contracts, this process must terminate.) Workers then self-select from among the contracts (if any) that are left. Otherwise, things are just as in the models of this chapter.

What can you say about the subgame perfect equilibria of this game? Can you give a precise characterization? If not, can you give partial characterizations? Can you describe at least one equilibrium? (There are many proper subgames here, and you may find things easier if you look for Nash equilibria where workers play sequentially rational strategies. That is, don't worry about subgame perfection for the firms.)

■ 8. Besides education as a signal, the other classic application of these ideas is the insurance market. The following simple story is usually told. Imagine an individual who is subject to a random loss. Specifically, the individual may have Y_1 with which to consume, and she may have only $Y_2 < Y_1$. This individual is strictly risk averse, with von Neumann-Morgenstern utility function U defined on the amounts of money she has

to consume. There is a probability π_i that the individual will have income Y_2 only (and a probability $1 - \pi_i$ that she will have income Y_1), where the subscript i stands for either H or L, mnemonics for high and low risk. We have $\pi_H > \pi_L$. The individual knows whether she is a high- or low-risk individual.

Several insurance companies offer insurance against the bad outcome for this consumer. An insurance contract is very simple: It specifies amounts y_1 and y_2 that the individual is left with if she would otherwise be consuming Y_1 or Y_2 . That is, you can think of $y_1 = Y_1 - P$, where P is the insurance premium, and $y_2 = Y_2 - P + B$, where B is a benefit paid in case the individual would otherwise be left with only Y_2 . The insurance companies are competitive and risk neutral. This implies that contracts that are offered and taken in equilibrium on average break even, or $\pi(Y_2 - y_2) + (1 - \pi)(Y_1 - y_1) = 0$, where π is the chance that the consumer would otherwise be left with Y_2 . The insurance companies do not know what type the consumer is, but have equilibrium conjectures based on the type of contract that the consumer proposes or takes. Insurance firms begin with the prior assessment that the consumer is high risk with probability $1/2$.

Adapt first Spence and then Rothschild and Stiglitz to this setting.

The key is to get the right "picture," and we will give you some assistance here. Think of the "commodity space" (replacing (w, e) space) as the space of pairs (y_1, y_2) where y_1 is the amount of money the consumer has to use in the contingency where she would otherwise have Y_1 and y_2 is her wealth in the contingency where she would otherwise have Y_2 . Begin by locating the consumer's endowment (Y_1, Y_2) . Then draw in indifference curves for the consumer in this space. Note carefully, the indifference curves are different if the consumer is high-risk or low-. It is useful to compute the slope of the consumer's indifference curve at a point (y_1, y_2) where $y_1 = y_2$. Finally, find the locus of zero-profit contracts (y_1, y_2) for the insurance firm if it knows that it is dealing with a high-risk client, with a low-risk client, and with a client who is high-risk with probability $1/2$. (Our offer of this assistance is not entirely altruistic. If you get stuck, or you succumb to laziness, the answer can be found at the start of the next chapter.)

■ 9. In a particular population, everyone runs the risk of losing \$1,000 randomly. Each person's random event (which determines whether the person loses the \$1,000 or not) is independent of everyone else's, and the chance that any single individual will lose the \$1,000 depends on the

individual; 90% of the population will lose the \$1,000 with probability .1, while the other 10% will lose the \$1,000 with probability .6. Each individual knows her own probability of loss, but no one knows the probability of loss of anyone else. Everyone in this population has the same von Neumann-Morgenstern utility function for this particular gamble, (either lose \$1,000 or lose \$0), given by $u(x) = 1 - e^{-\lambda x}$, where x is the net of this gamble.

A regulated insurance company stands ready to offer insurance for this gamble. If an individual pays a premium P , then the individual is insured against the loss of \$1,000. In terms of the net, the individual trades a gamble where the two possible prizes are $x = 0$ and $x = -1,000$ for a sure thing where $x = -P$. Because this insurance company is regulated, it must set the premium so that its expected net profit is zero.

(a) If $\lambda = .002$, is there a pooling equilibrium for this economy? If $\lambda = .0005$, is there a pooling equilibrium? For what values of λ is there a pooling equilibrium? (Answer the last question only if you have access to a computer or a programmable calculator.) As part of your answer to this, you must say what constitutes a pooling equilibrium.

(b) Suppose $\lambda = .001$, and individuals can undergo an examination that with some chance of error identifies whether the individual has a chance of .1 of losing the \$1,000 or a chance of .6. Specifically, this exam has two outcomes: H (for high risk) and L. If the individual is the sort who will lose the \$1,000 with probability .1, then this test gives the outcome L with probability p and the outcome H with probability $1 - p$. If the individual will lose the \$1,000 with probability .6 then the outcomes are H with probability p and L with probability $1 - p$. It costs \$10 to administer this test. The regulatory body says that the insurance company must charge any consumer who wants the test a nonrefundable \$10 (consumers may only take the test once), and the company must then break even on each and every contract that it offers. (That is, one contract may not be allowed to cross-subsidize another.) The insurance company may make the premium dependent on the results of the test if taken. Will this economy admit a screening equilibrium? What is the nature of that equilibrium? For what values of p will such an equilibrium exist?

Note: In both halves of this problem, do not assume competition among insurance companies. The single insurance company is subject to regulation, as noted above. If you want to think about it, you might consider what would happen if there was a competitive insurance industry.

■ 10. Return to section 13.2 and the story of entry deterrence told there. If you didn't understand the equilibria that were described then, you should go back to them now. (Can you now find all the equilibria?)

In the game theoretic analysis of the Spence model given here, we saw that pooling equilibria are destroyed if we employ equilibrium dominance criteria. This is not true in the pooling equilibrium we sketched in section 13.2. Why not? (Hint: When you summarize your answer, begin with the following line: "In the model of this chapter, if there is a pool, the high-ability worker suffers for it. But...")

■ 11. We have now discussed problems of adverse selection and problems of moral hazard. In this problem, you may try your hand at a situation with both at the same time. If you find this problem too hard, wait until the end of the next chapter, when you should find it very easy. Consider a situation just like the salesman problem in section 16.2, with one additional fillip. Specifically, the salesman will be of either high or low ability. To be exact: You are going to hire a salesman to try to make a particular sale for you. You are risk neutral in all this. He is risk averse, with von Neumann-Morgenstern utility function $\sqrt{x} - a$, where x is the amount you pay him, and a is an index of the level of effort he exerts. He is either of good ability or of bad ability, each with probability .5. In parts (a) and (b) you will know his level of ability; in later parts you will not. He will exert either a high level of effort or a low, and you will be unable to observe which. (Note: Good and bad pertain to ability, and high and low pertain to effort.) Regardless of his level of ability, he has reservation utility level 5. And if he puts in a high level of effort, a in his utility function is 1; $a = 0$ for a low level of effort.

He either will make the sale or he won't. The sale, gross of your payment to him, will net you \$100 if he makes it. The chances that he makes the sale depend both on his level of ability and on his level of effort. Specifically:

If he is good and gives high effort, he makes the sale with probability .9.

If he is good and gives low effort, he makes the sale with probability .6.

If he is bad and gives high effort, he makes the sale with probability .5.

If he is bad and gives low effort, he makes the sale with probability .3.

(This is a world in which ability counts more than effort.)

Although it doesn't make much sense in the context I have outlined, assume that you are only allowed to deal with one salesman. You may offer

him a contract, which he will either accept or reject; if he rejects, you net zero. Any time that the salesman is indifferent, he will elect the option that is more favorable to you. (That is, you don't have to worry about ties.)

(a) Suppose you are able to observe his ability, and you find that the salesman you are dealing with has bad ability. What, from your point of view, is the optimal contract to offer him? Will he give high or low effort under this contract? What will be your profits?

(b) Suppose you are able to observe his ability, and you find that the salesman you are dealing with has good ability. What, from your point of view, is the optimal contract to offer to him? Will he give high or low effort under this contract? What will be your profits?

(c) Now suppose that you are unable to observe the ability of the salesman you are dealing with. If you offer him a choice from the contracts you identified in parts (a) and (b), what will be the outcome for you? Why is this? Be very clear and explicit as to what you expect to happen in this case.

(d) You are unable to observe the ability of the salesman that you are dealing with, and you may either offer him a pair of contracts from which he can choose, or a single contract, that he may elect to take or not. (If you offer a pair of contracts, he can refuse employment altogether as well, although remember that when he is indifferent, he is presumed to do what is in your interests.) I want you to produce an offer to him that does better than offering him the pair of contracts from parts (a) and (b). Say what the contract(s) is (are), and what will be the outcome of your offer. You should produce the best contract(s) you can — ideally, you should devise the optimal contract(s) to offer him, and you should prove that this is the optimal offering.

■ 12. At the start of this chapter we used Gresham's law to motivate Akerlof's lemons model and problems of adverse selection. But this motivation is not entirely appropriate, at least for the way we described Gresham's law. Why not? How can you make Gresham's law appropriate? (For the second question, begin your answer with "Suppose some of the people in this economy are pathologically honest....")

chapter eighteen

The revelation principle and mechanism design

In chapter 16, we saw how to design optimal incentive contracts for a principal who wishes to get a single agent to take some action. We hinted there, both in the text and in the problems, at techniques for the design of optimal contracts when the agent possesses private information and when there are multiple agents.

In this chapter, we look at a grab bag of situations in which these techniques are employed. The specific applications that we will examine, insurance contracts when there is adverse selection (and, more generally, optimal nonlinear pricing), optimal procurement, and elicitation of preferences for the provision of a public good, are all interesting in their own right, and we carry through the first level of analysis for all. But the point of this chapter is the general mode of analysis that is employed and, in particular, the progression from contracts designed for a single party to the revelation principle and contracts and mechanisms designed for many parties.

18.1. Optimal contracts designed for a single party

At the end of the last chapter, we asked what would happen in an insurance setting with adverse selection if the insurance industry instead of being competitive is either a monopoly or is cartelized so effectively that it acts like a monopoly. In chapter 8 (problem 4), we posed the problem of finding an optimal nonlinear pricing scheme in the context of a manufacturing monopolist selling to retail monopolists. In this section we see how to solve these two problems.

Profit-maximizing insurance contracts with adverse selection

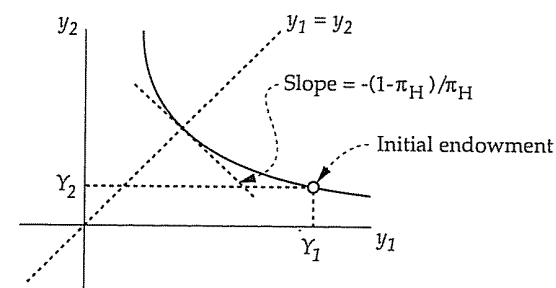
To begin, we develop a context within which the question of profit-maximizing insurance can be studied.¹ Imagine an individual who will

¹ This analysis is taken from Stiglitz (1977).

receive an uncertain amount of money with which to consume. Specifically, the individual may receive the amount Y_1 , or she may receive only $Y_2 < Y_1$. Which of these two prevails is determined at a later date. This individual is strictly risk averse, with strictly increasing and strictly concave von Neumann-Morgenstern utility function U defined on the amounts of money she has to consume. There is a probability π_i that the individual will have endowment Y_2 only and a probability $1 - \pi_i$ that she will have endowment Y_1 , where the subscript i stands for either H or L, mnemonics for high and low risk. We assume $\pi_H > \pi_L$. The individual knows whether she is a high- or low-risk individual.

Because she is subject to endowment risk and she is risk averse, this consumer might seek to insure herself against a low level of endowment. In essence, she will give up some of her endowment in the high-endowment state of nature in return for more in the low-endowment state. Suppose there is a single insurance company with which the consumer can deal. An insurance contract between the insurance company and the consumer specifies two numbers, y_1 and y_2 , where y_k is the total resources left to the consumer if her endowment is Y_k , for $k = 1, 2$. You can think of $y_1 = Y_1 - P$, where P is the insurance premium, and $y_2 = Y_2 - P + B$, where B is a benefit paid if the individual would otherwise be left with only Y_2 . The insurance company, then, will have "profits" $Y_1 - y_1$ in the high-endowment state and $Y_2 - y_2$ in the low-endowment state; we put "profits" in quotes because one or the other of these quantities will presumably be negative. (In fact, it is reasonable to presume that $Y_2 - y_2$ will be negative; the insurance company pays out in the low-endowment state.) The insurance company does not know whether the consumer is high or low risk and begins with the prior assessment that the consumer is high risk with probability ρ . (What sort of contract(s) will the insurance company offer to this consumer in order to make the largest possible expected profit?

It may help to draw some pictures of this situation. The "commodity space" for the consumer is the set of all pairs (y_1, y_2) , with the interpretation that y_1 is the amount the consumer has to spend in the state of nature where she would (without the benefit of insurance) consume Y_1 , and y_2 is the amount she has to spend in the state of nature where her initial endowment is Y_2 . We can draw indifference curves for the consumer in (y_1, y_2) space; these are curves of the form $(1 - \pi_i)U(y_1) + \pi_i U(y_2) = k$ for constant k where π_i is the consumer's probability of having a low level of endowment. In figure 18.1 we draw the indifference curve of a high-risk consumer through the endowment point (Y_1, Y_2) .



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Figure 18.1. A high-risk consumer's indifference curve.

Fix the constant k , and think of the curve as determining y_2 as a function of y_1 . That is, define a function $y_2(y_1)$ by the implicit relation

$$(1 - \pi_i)U(y_1) + \pi_i U(y_2(y_1)) = k.$$

Letting V be the inverse of U , this is

$$y_2(y_1) = V\left(\frac{k - (1 - \pi_i)U(y_1)}{\pi_i}\right).$$

(We should be worried whether $[k - (1 - \pi_i)U(y_1)]/\pi_i$ is in the domain of V , but we'll simply proceed on the assumption that it is, and we leave it to you to clean up the details.) Hence

$$\frac{dy_2}{dy_1} = -V'\left(\frac{k - (1 - \pi_i)U(y_1)}{\pi_i}\right)\left(\frac{1 - \pi_i}{\pi_i}\right)U'(y_1), \quad \text{or}$$

$$\frac{dy_2}{dy_1} = -V'(U(y_2))\left(\frac{1 - \pi_i}{\pi_i}\right)U'(y_1) \quad \text{for} \quad y_2 = V\left(\frac{k - (1 - \pi_i)U(y_1)}{\pi_i}\right).$$

This looks like (and is) a mess, but from it we can harvest two facts that are useful in drawing pictures.

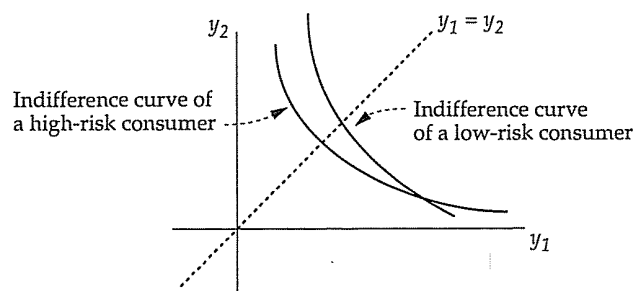


Figure 18.2. A pair of indifference curves, one for each type.

(1) If $y_2(y_1) = y_1$, then the slope of the indifference curve through (y_1, y_2) equals $-(1 - \pi_i)/\pi_i$. (This follows because $V'(U(x)) = 1/U'(x)$.)

Consider (superimposing) indifference curves of the consumer for the two cases where the consumer is high and low risk. In particular, consider the indifference curves of the consumer through some point (y_1, y_2) . (2) The indifference curve of the low-risk consumer is steeper than that of the high-risk consumer. Put differently, if we start from a point (y_1, y_2) and decrease y_1 by a little bit, it takes a smaller increase in y_2 to compensate the high-risk consumer than to compensate the low-risk consumer. The intuition for this should be clear; the high-risk consumer has a greater chance of having the low-endowment state prevail, hence additional income in that state is worth more to her. We also see this algebraically, since the slopes are the same except for the scale factors $(1 - \pi_i)/\pi_i$; this factor is lower for the high-risk consumer than for the low-risk.

Indeed, we have something stronger, namely that the ratio of the slopes of the indifference curves through any point is independent of the point. But we won't need this stronger result. All we need is the "single-crossing" condition for any pair of indifference curves, one for a high-risk consumer and one for a low-, with the crossing as shown in figure 18.2.

Consider next (iso-expected) profit lines for the firm. Suppose the firm sells a contract to the consumer that leaves the consumer at the point (y_1, y_2) . The firm is left with profits $Y_1 - y_1$ in the high-endowment state and $Y_2 - y_2$ in the low-endowment state, hence the firm's expected profits are $(1 - \pi)(Y_1 - y_1) + \pi(Y_2 - y_2)$, where π is the firm's overall assessment as

18.1. Optimal contracts designed for a single party

to how likely it is that the consumer will have a low level of endowment. Iso-expected profit lines for the firm are given by

$$(1 - \pi)(Y_1 - y_1) + \pi(Y_2 - y_2) = k'$$

for some constant k' , and a bit of algebra shows that the slope of these (viewed with y_2 on the ordinate and y_1 on the abscissa) is $-(1 - \pi)/\pi$.

Suppose that the firm knows it is dealing with a high-risk consumer, so it assesses $\pi = \pi_H$. What contract does it wish to offer to the consumer in order to (maximize its expected profits? The picture is as in figure 18.3. The consumer isn't going to settle for any contract that leaves her worse off than at her endowment point. So the firm is constrained to offer her a contract in the shaded area, if it wishes to do business. Its highest expected profits are at the point along her indifference curve where $y_2 = y_1$. (We have discovered yet again the fact, first discovered in chapter 5, that efficient risk sharing between a risk-averse and a risk-neutral party shifts all the risk onto the risk-neutral party.)

Similarly, if the firm knows it is dealing with a low-risk consumer, it offers the contract with $y_2 = y_1$ along the low-risk consumer's indifference

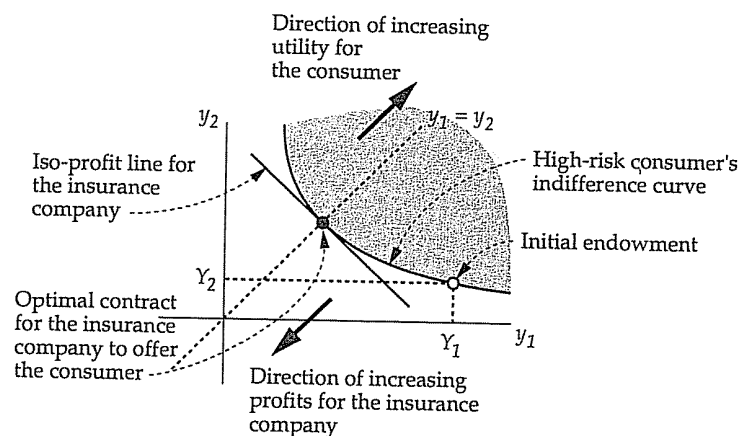


Figure 18.3. The optimal contract to offer a high-risk consumer.

curve through the endowment point. Since the low-risk consumer has indifference curves that are more steeply sloped, this will be at a point that is "higher up" the line $y_2 = y_1$ than is the point for high-risk consumers. (Put differently, this is because the low-risk consumer is less willing to buy "expensive" insurance.)

Either from the picture or algebraically, it is straightforward to prove the following further result:

Lemma. Suppose the firm offers a contract (y_1, y_2) that is accepted by some type of consumer. If we change that contract in a way that keeps the utility of this type of consumer constant and we move in the direction of full insurance, $y_1 = y_2$, then the expected profits of the insurance firm (from this type of consumer accepting the contract) increase.

In other words, in figure 18.3, not only is the highest level of expected profits for the firm along the high-risk consumer's indifference curve to be found along the line $y_1 = y_2$, but expected profits are increasing on each side as the contract is moved along this indifference curve towards this line.

All this, it should be noted, has been done with the implicit assumption that the firm can make a take-it-or-leave-it offer to the consumer. The firm offers to trade some pair (y_1, y_2) for the consumer's initial endowment or, equivalently, the firm says that it will charge a premium $Y_1 - y_1$ in exchange for a guarantee that the consumer will have y_1 to consume when times are tough. It is possible, though, that the insurance company cannot make offers of this sort; perhaps it must name a premium P and a low-endowment state payout B , and the consumer can choose how much of this insurance to buy. We leave it to you to discover how this would change the story we are telling. We will continue under the hypothesis that the firm is able to make take-it-or-leave-it offers.

We come back to the question with which we began. The insurance company can't tell whether the consumer is high risk or low. What should it do in this case? It clearly won't do to offer the optimal contracts for the two types of consumer and expect the consumer to select the one that is "right" for her. If the firm does this, the consumer, whether high risk or low, is going to choose the contract optimally designed for the low-risk consumer, since this gives a higher level of income in both states. The insurance firm could offer the optimal contract for the high-risk consumer alone, but then the low-risk consumer will refuse to buy.

Or consider the scheme depicted in figure 18.4. We have in mind that the firm offers the two contracts shown as the heavy dot and square

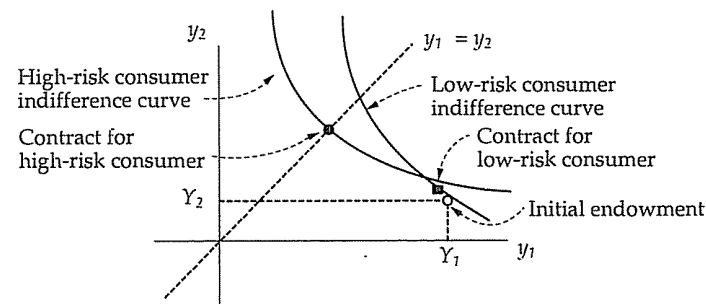


Figure 18.4. A possible pair of contracts.

and allows the consumer to pick one. The contract depicted by a dot represents complete insurance; the square represents only partial insurance. We have also drawn in the indifference curves of the high-risk consumer through the dot and low-risk consumer through the square. From this you can see that given a choice between the dot, the square, or no contract at all (the open circle) the high-risk consumer will choose the dot, and the low-risk consumer will choose the square. In this arrangement, the firm provides complete insurance to the high-risk consumer, although on somewhat better terms for the consumer than if the firm knew it was dealing with the high-risk consumer. It provides partial insurance for the low-risk consumer as well. Perhaps some scheme of this sort is optimal for the firm.

We can solve the firm's problem analytically as follows. We imagine that the firm offers two contracts, (y_1^H, y_2^H) and (y_1^L, y_2^L) . The contract (y_1^H, y_2^H) is intended for the high-risk consumer and the contract (y_1^L, y_2^L) for the low-. If we assume that the consumer chooses the contract intended for her, the firm's expected profits are

$$\rho[(1 - \pi_H)(Y_1 - y_1^H) + \pi_H(Y_2 - y_2^H)] + (1 - \rho)[(1 - \pi_L)(Y_1 - y_1^L) + \pi_L(Y_2 - y_2^L)].$$

This is what the firm seeks to maximize. But we have to make sure that the consumer picks the contract that is intended for her. There are four constraints to add: The high-risk consumer must prefer (y_1^H, y_2^H) to her endowment (Y_1, Y_2) . The high-risk consumer must prefer (y_1^H, y_2^H) to (y_1^L, y_2^L) . The low-risk consumer must prefer (y_1^L, y_2^L) to her endowment (Y_1, Y_2) .

And the low-risk consumer must prefer (y_1^L, y_2^L) to (y_1^H, y_2^H) . In order, these four constraints are:

$$(1 - \pi_H)U(y_1^H) + \pi_H U(y_2^H) \geq (1 - \pi_H)U(Y_1) + \pi_H U(Y_2); \quad (PH)$$

$$(1 - \pi_H)U(y_1^H) + \pi_H U(y_2^H) \geq (1 - \pi_H)U(y_1^L) + \pi_H U(y_2^L); \quad (IH)$$

$$(1 - \pi_L)U(y_1^L) + \pi_L U(y_2^L) \geq (1 - \pi_L)U(Y_1) + \pi_L U(Y_2); \quad (PL)$$

$$(1 - \pi_L)U(y_1^L) + \pi_L U(y_2^L) \geq (1 - \pi_L)U(y_1^H) + \pi_L U(y_2^H). \quad (IL)$$

The names given to the four constraints are mnemonics for: The Participation constraint for the High-risk type; the Incentive constraint for the High-risk type; and similar names for the Low-risk types.

So we have the problem: Choose the four variables y_1^H, y_2^H, y_1^L , and y_2^L to maximize the objective function shown, subject to these four constraints.

You may be a bit queasy about this formulation on three grounds. The first two are: We seem to be assuming that the firm offers different contracts for the two types of consumer. Why can't the firm choose to pool the two types by offering a single contract? And we seem to be assuming that the firm serves both types. Is it possible that the firm will wish to offer a contract that only one type and not the other will accept, leaving the other type uninsured? The formulation given allows for each of these two possibilities. We have not precluded finding a solution where $y_1^H = y_1^L$ and $y_2^H = y_2^L$. If this turns out to be the solution, the two types are pooled. And we haven't precluded finding a solution where $(y_1^H, y_2^H) = (Y_1, Y_2)$ or where $(y_1^L, y_2^L) = (Y_1, Y_2)$. In the first of these cases, the firm is "offering" the high-risk consumer her original endowment, which is equivalent to leaving this type to fend for herself. If we find a solution where the second case prevails, it is the low-risk consumer who is not being given any insurance at all.²

The third reason for unease is more substantial. We have blithely assumed that the optimal "mechanism" for the insurance firm to use is to offer the consumer a menu of two contracts, one intended for the consumer if she is low risk, and one intended for her if she is high risk. Perhaps some more complex scheme for marketing insurance will make more profits for the firm.

If the consumer anticipates all the things the firm will do in marketing insurance, and if she responds to these things rationally, then our menu of two contracts is without loss of generality. The argument runs as follows:

² See the remarks in the next subsection on mixed moral hazard and adverse selection problems for another way to do this.

Whatever scheme the insurance company plans to adopt, it will end with the consumer taking some contract or another. A contract will be taken by the low-risk consumer, and a contract will be taken by the high.^a These contracts could be the same, they could be different, and either or both could be the endowment point. Now for the crux of the argument. If the consumer anticipates all this, she must prefer the low-risk contract to the high-risk contract if she is low risk and she must prefer the high-risk contract to the low-risk contract if she is high risk. The reason is that if she is, say, high risk, and she prefers the low-risk contract to the high-risk contract, she could obtain the low-risk contract simply by mimicking the actions that she would take if she were low risk. The insurance company can't tell the difference. Moreover, the contract that she winds up with must be as good as the endowment point for her, as long as she retains the right to say no. So any marketing scheme, however complex, comes down to the final contracts that the consumer might have, and the firm can do no better than to offer the menu of these contracts at the outset and let the consumer choose.

We stress the proviso that the consumer must rationally anticipate the marketing scheme the firm will employ. The firm might do better than to offer a menu if the consumer does not anticipate all the things the firm will do along the way through some scheme. The consumer might not anticipate that certain options will be made available to her if she makes certain choices early on; we will return to this in a couple of subsections when we discuss the credibility problem facing the insurance firm at the optimal contract. The consumer might be misled into thinking that certain options will be offered to her that will not appear, and so on. Without meaning to cast aspersions at the honorable profession of marketing, it sometimes seems that marketing schemes are devised to confound the consumer. *A confoundable consumer might be confounded by a complex marketing mechanism and not by a simple choice from a menu.* Insofar as this is so, the maximization problem above may not find the optimal scheme for the insurance company. But we assume as part of the basis for this form (and most other forms) of economic analysis that consumers are not confoundable. The consumer can work through the choices she will be offered, and she responds optimally. Given this, the two-choice menu scheme (one for

^a We should be careful here. Perhaps the marketing mechanism employed by the insurance company will result in a random assignment of contracts to the consumer; the consumer, if she is high risk, gets contract A with some probability, contract B with some other, and so on. We will ignore mechanisms with random outcomes here, for the reason that they are never optimal in this setting, but a more careful analysis would allow for them and then derive the fact that they are never optimal.

each of the two types of consumer) will give the optimal scheme for the insurance firm.

Having settled the relevance of the constrained optimization problem given, we could dive into the problem algebraically, looking at first-order equations and complementary slackness conditions. Instead, we will argue graphically.

But before moving to graphical analysis, a remark is in order about trying algebra. The problem is a bit easier to analyze if we first make a change of variables. Let the decision variables be $u_k^H = U(y_k^H)$ and $u_k^L = U(y_k^L)$, for $k = 1, 2$. Then the objective function becomes

$$\rho[(1 - \pi_H)(Y_1 - V(u_1^H)) + \pi_H(Y_2 - V(u_2^H))] \\ + (1 - \rho)[(1 - \pi_L)(Y_1 - V(u_1^L)) + \pi_L(Y_2 - V(u_2^L))],$$

where V is the inverse of U . Note that if U is concave, V is convex, so $-V$ is concave, and we will be seeking to maximize a concave function of our variables. And the four constraints become linear; for example, (IH) becomes

$$(1 - \pi_H)u_1^H + \pi_H u_2^H \geq (1 - \pi_H)u_1^L + \pi_H u_2^L.$$

Note that since, with this change of variables, the objective function (to be maximized) is concave and the constraints are linear, we know that solution of the first-order and complementary slackness conditions (together with feasibility) are necessary and sufficient for a solution. The main advantage of this change of variables, though, is that the algebra is usually simpler. In particular, if you are diligent, you may wish to replicate the graphical analysis to follow algebraically, using the first-order and complementary slackness conditions. Having linear constraints will be a blessing in this activity.

Proposition 1. *At the solution of this problem, the constraints (PL) and (IH) will be binding, and the high-risk contract will be a full-insurance contract; i.e., $y_1^H = y_2^H$.*

The proof of this result will be given in a number of steps. Throughout, we imagine that $\{(y_1^H, y_2^H), (y_1^L, y_2^L)\}$ is a solution of the problem. Whenever we draw pictures, we use an open circle for the endowment point, a filled in circle for (y_1^H, y_2^H) , and a filled in square for (y_1^L, y_2^L) .

Step 1. Consider figure 18.5(a). We have drawn there the low-risk contract (y_1^L, y_2^L) , and through this point we have drawn indifference curves for the high- and low-risk types of consumer. The high-risk contract (y_1^H, y_2^H) lies within the shaded "wedge."

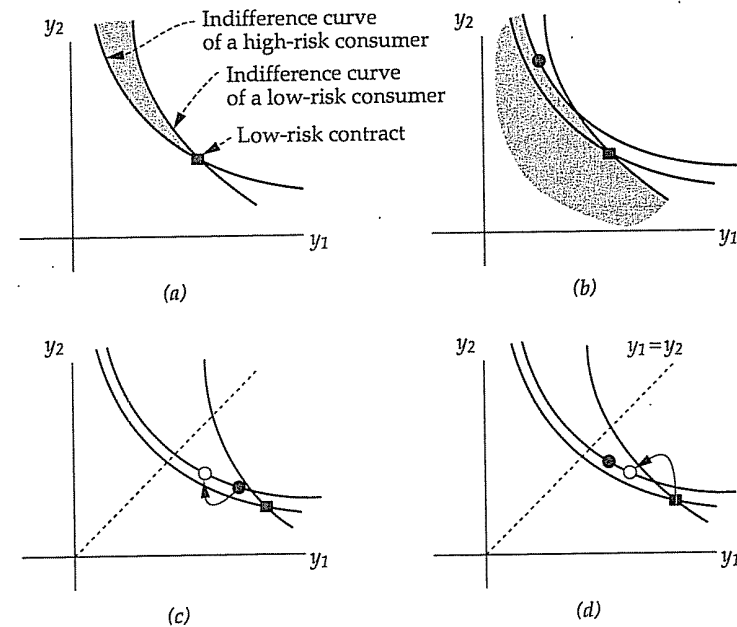


Figure 18.5, part 1. Graphical analysis of profit-maximizing insurance. In (a), we show the "wedge" relative to (y_1^L, y_2^L) within which (y_1^H, y_2^H) must lie. In (b) we show the region in which (Y_1, Y_2) must lie relative to (y_1^L, y_2^L) and (y_1^H, y_2^H) . In (c) moving (y_1^H, y_2^H) to the point (Y_1, Y_2) improves expected profits while maintaining feasibility; in (d) the same is true for (y_1^L, y_2^L) .

This follows immediately from the constraints (IH) and (IL). To satisfy (IH), (y_1^H, y_2^H) must lie on or above the high-risk indifference curve through (y_1^L, y_2^L) . To satisfy (IL), (y_1^H, y_2^H) must lie on or below the low-risk indifference curve through (y_1^L, y_2^L) . The intersection of these two regions defines the wedge.

Step 2. The constraint (PL) binds. In particular, (Y_1, Y_2) must lie along the low-risk indifference curve through (y_1^L, y_2^L) to the right of (y_1^L, y_2^L) .

To see this, consider figure 18.5(b). We have drawn there (y_1^L, y_2^L) , (y_1^H, y_2^H) , and indifference curves for both types through (y_1^L, y_2^L) and for the high-risk type through (y_1^H, y_2^H) . We note first that (Y_1, Y_2) must lie in the shaded region of 18.5(b) in order to satisfy (PH) and (PL). Also, it is

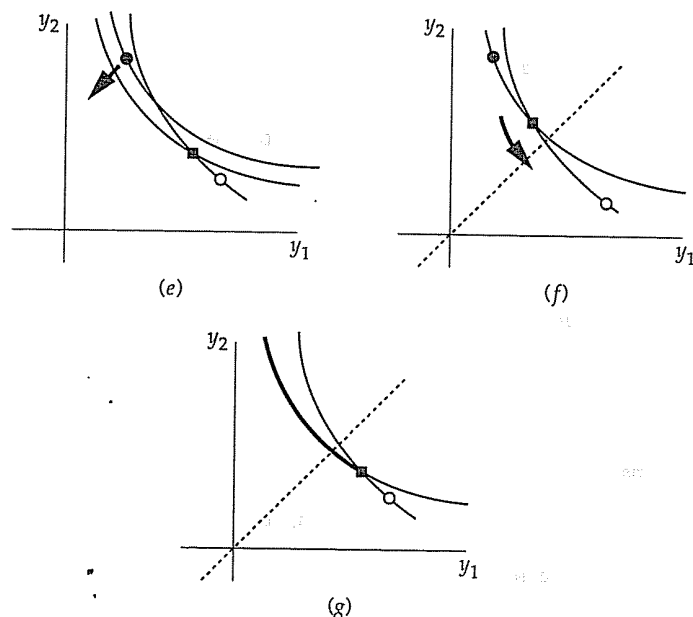


Figure 18.5, part 2. Graphical analysis of profit-maximizing insurance.

Thus (Y_1, Y_2) must lie as shown in (e). But in (e) there is the problem that (IH) doesn't bind, and movement of (y_1^H, y_2^H) in the direction indicated improves profits. In (f) is a situation where (y_1^L, y_2^L) involves overinsurance, in which case movement in the direction indicated improves profits. And in (g), the portion of the high-risk indifference curve through (y_1^H, y_2^H) lying to the left of (y_1^H, y_2^H) is feasible, and full insurance is the point that maximizes expected profits.

impossible that both (PL) and (PH) are slack, for if this were the case, we could lower the payments made in the two contracts somewhat without affecting feasibility (taking care so that (IH) and (IL) continue to hold), which would certainly increase the firm's expected profits. Hence (Y_1, Y_2) must lie along the boundary between the shaded and unshaded regions in figure 18.5(b).

Suppose (Y_1, Y_2) lies to the left of (y_1^H, y_2^H) , necessarily along the high-risk indifference curve through (y_1^H, y_2^H) , as in figure 18.5(c). Then all the

constraints continue to be satisfied if we move (y_1^H, y_2^H) to the point (Y_1, Y_2) , and profits increase because $Y_1 > Y_2$.

Suppose (Y_1, Y_2) lies at or to the right of (y_1^H, y_2^H) and to the left of (y_1^L, y_2^L) . One case is shown in figure 18.5(d). We claim that if we move (y_1^L, y_2^L) to the point (Y_1, Y_2) , expected profits increase and all the constraints continue to be satisfied. To see that expected profits increase, note that this shift involves movement along the low-risk indifference curve in the direction of full insurance (since $Y_1 > Y_2$) and (in the case depicted) a lowering of both payments. You can see by inspection that the constraints all continue to hold.

Hence (Y_1, Y_2) must lie along the low-risk indifference curve through (y_1^L, y_2^L) to the right of (y_1^L, y_2^L) .

Step 3. The constraint (IH) must bind. By steps 1 and 2, the picture must be as in figure 18.5(e). But if (IH) doesn't bind, as is depicted in figure 18.5(e), then movement of (y_1^H, y_2^H) in any direction down and to the left, staying within the wedge, doesn't compromise feasibility and improves expected profits. The key is the positioning of (Y_1, Y_2) , which implies that as long as (y_1^H, y_2^H) stays within the wedge, (PH) is maintained.

Step 4. The contract (y_1^L, y_2^L) does not involve overinsurance; that is, $y_1^L \geq y_2^L$. To see this, consider figure 18.5(f), where a situation of overinsurance is depicted. If (y_1^L, y_2^L) is moved along the low-risk indifference curve in the direction of the arrow, towards full insurance, feasibility is maintained (the constraints on y^H are actually loosened), and expected profits increase by the lemma.

Step 5. The contract (y_1^H, y_2^H) involves full insurance. By steps 1 through 4, the picture must be as in figure 18.5(g), where (y_1^H, y_2^H) lies along the high-risk indifference curve to the left of (y_1^L, y_2^L) and the full-insurance line lies on or to the left of (y_1^L, y_2^L) . All points along the high-risk indifference curve to the left of (y_1^L, y_2^L) maintain feasibility and, by the lemma, we know that full insurance maximizes expected profits along this curve.

From the proposition, we see that the problem is essentially one dimensional. Consider figure 18.6. The high-risk contract (y_1^H, y_2^H) must involve full insurance and must lie between the high-risk and low-risk indifference curves through the endowment. Once the location of (y_1^H, y_2^H) is selected, we know that (IH) and (PL) must be binding, so the location of (y_1^L, y_2^L) is forced to be at the intersection of the high-risk indifference curve through (y_1^H, y_2^H) and the low-risk indifference curve through (Y_1, Y_2) . When we put (y_1^H, y_2^H) at the first-best location for a high-risk consumer, we are forced to put $(y_1^L, y_2^L) = (Y_1, Y_2)$. Essentially, this involves first-

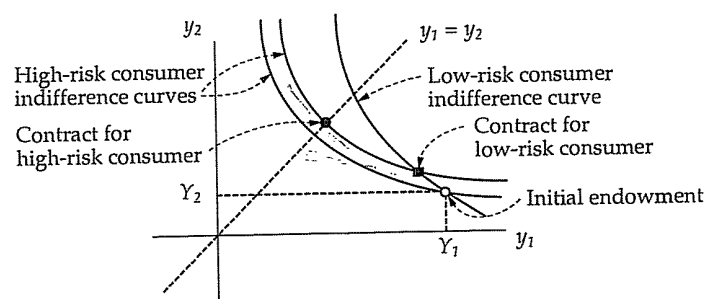


Figure 18.6. The picture for profit-maximizing insurance.

best insurance for a high-risk consumer and no insurance for a low-risk consumer. When we put (y_1^H, y_2^H) at the first-best location for a low-risk consumer, $(y_1^L, y_2^L) = (y_1^H, y_2^H)$ is forced. This corresponds to full insurance for both types, at terms that are best (from the perspective of the firm) for low-risk types. Otherwise, we get full insurance for high-risk types and partial insurance for low-risk types. Note that the only possibility of pooling is at $(y_1^L, y_2^L) = (y_1^H, y_2^H)$ and both equal to the first-best contract for low-risk types.

Now think of varying ρ parametrically, beginning with $\rho = 1$ (the consumer is sure to be the high-risk type) and decreasing ρ to zero. When $\rho = 1$, it is clear that the optimum is to provide first-best insurance for the high-risk consumer. When $\rho = 0$, it is clear that the optimum is to provide first-best insurance for the low-risk consumer. It can be shown that as ρ decreases, the contract (y_1^H, y_2^H) moves up the full-insurance line. So all that remains to be answered is the question, For $\rho \in (0, 1)$ (that is, there is strictly positive probability that the consumer is either type), can we get one of the "corner" solutions, or do we necessarily get one of the interior solutions? This question is of particular interest for the case ρ near 0. As we noted above, in this context we will have a pooling contract only at terms that involve complete insurance at the first-best contract for the low-risk types. If we knew that, for any $\rho > 0$, this is not the profit-maximizing contract (i.e., it is optimal only for $\rho = 0$), we would know that pooling does not occur. With this as a teaser, the analysis is left to you.

Variations, extensions, and complications

Besides being of independent interest, the analysis just completed illustrates how to formulate the problem of designing an optimal contract

for a single party in a situation where the party has private information. The party is presented with a menu of options, one option for every possible piece of private information that the party might possess, where the items on the menu are constrained so that for each piece of private information the party self-selects the item on the menu that is intended for "her." In the literature, you will find reference to various types of party, as in "the low-risk type" or "the high-risk type"; each piece of private information that the party might possess corresponds to a different type.

In the analysis above, the criterion maximized was profit of the insurance company. But one needn't restrict to profits. If we think of the insurance company as being regulated, then the regulatory body might wish to maximize some sort of weighted sum of the consumer's expected utilities. Or, taking shareholders into account, we might consider maximizing some criterion that puts positive weight on the firm's profits. (If the firm is regulated so that it must achieve some minimal return on its capital, we could include that as a constraint.)

Another variation concerns problems that combine moral hazard with adverse selection. Suppose the party for whom the contract is to be designed must take an action that affects the welfare of the party designing the contract. Suppose this action is taken under circumstances of moral hazard; i.e., the contract cannot (enforceably) specify what action the party will take. For example, in life insurance, the insuree might take better or worse care of herself, affecting the odds of early demise, and the insurance company cannot enforceably include in an insurance policy rules about the insuree's diet or exercise regime. The insurance company would then wish to design a menu contract that not only caused the insuree to identify herself on the basis of her health, but that also provided her with an incentive to take care of herself. A combination of techniques from this chapter and chapter 16 can be employed. Suppose there are T types of consumer, indexed by $t = 1, \dots, T$, and each type of consumer can take any action a from some finite set A . Imagine some selection of one action for each type, a function $\alpha : \{1, \dots, T\} \rightarrow A$. We can ask, Can the firm design a menu of contracts, $c_1, \dots, c_T$, such that a consumer of type t will select contract c_t and then choose action $\alpha(t)$? If so, what is the optimal (say, profit-maximizing) menu of contracts that implements α in this sense? This is a more complex problem only in that there are more constraints. For each type t , we have one constraint less than there are pairs $(t', a') \in \{1, \dots, T\} \times A$. If the individual is of type t , she must prefer to choose the contract option c_t and take action $\alpha(t)$ to choosing any other $c_{t'}$ and taking action a' , for any $t' \neq t$ or $a' \neq \alpha(t)$. Problem 11 of the previous chapter is attacked in this manner.

In essence, we did something along these lines in the example of this section. The actions available to the insuree were to choose a contract or to refuse all the contracts offered, and the (PH) and (PL) constraints were there to ensure that the agent took the desired action, namely signing a contract. (The option of choosing a contract but then not signing leaves the consumer at her endowment point.) We noted that we could mimic the option of not signing by including a contract that specified that the consumer would remain at her endowment; alternatively, we could use a structure of the sort just described, where we would find in turn the optimal expected profits given that both types sign, that the high-risk signs but not the low, that the low-risk signs but not the high, and that neither sign.

As we noted in the specific case of insurance contracts, our formulation assumes that the party designing the contract can make a take-it-or-leave-it offer to the second party. Because of the possibility of resale or purchase from multiple vendors, the seller of a particular good or service may be unable to make this sort of offer.

For example, imagine that insurance companies are very effectively cartelized (thus motivating our attention to profit-maximizing contracts). Imagine that the equilibrium has the low-risk consumer obtaining only partial insurance. We might then imagine that the high-risk consumer purchases partial insurance from a number of different firms, getting terms better than those she can get by buying complete insurance from one firm. (This is a good exercise for graphical analysis. Consider figure 18.6. If insurance policies are written to specify a premium and a payment in the low-endowment state, and the high-risk consumer buys two insurance policies of the sort intended for the low-risk consumer, what will be her levels of consumption? How does she feel about this combination vis à vis the full insurance contract that is intended for her? How do the insurance companies feel about this?) If the consumer can do this, it will wreak havoc with the insurance firms' supposedly profit-maximizing contracts. So insurance companies might put into their policies clauses about not being able to insure the same risk with two different carriers.³

In general, these sorts of considerations can restrict the menus that a firm can offer. Analysis along the lines we have indicated is possible, but it is often harder.

A final "generalization," and one that is immensely complicated, involves contract design by an informed party. Imagine a situation in which

³ If you know about the insurance business and, in particular, moral hazard problems associated with full or even overinsurance, you will see another reason for insurance companies to write such clauses into their policies.

a government agency wishes to procure some items from a manufacturer. The government agency may not know the cost structure facing the firm as well as the firm does, and so it may design a menu of contracts from which the firm is to choose. If the firm is a higher cost firm, it chooses a contract with a greater amount of cost sharing, say, while a lower cost firm might choose a contract with less cost sharing but a higher unit price. It is easy to imagine, in this case, that the government possesses information not held by the firm, say about the engineering costs that the particular item entails. Depending on what the government knows, the menu of contracts that would be optimal for the government to offer might change. But then the firm, seeing what menu it is offered, might learn something about what the government knows. In designing the menu of options to offer the firm, the government must decide how "inscrutable" it should be. This involves thinking through what inferences the firm will make as a function of the menu of options the firm is offered, and how those inferences will affect what options the firm chooses given its own private information.

This variation on the basic puzzle falls under the rubric of "mechanism design by an informed principal." It is a good deal more complicated than what we have described above, and we leave it to you to read about in the literature.

The problem of credibility

One complication of (or caveat to) the basic story given above should be noted especially well. Suppose that in our example the insurance company offers two contracts, one with complete insurance intended for the consumer if she is high risk and one with partial insurance intended for her if she is low risk. Suppose she takes the low-risk insurance contract. Since this is a partial insurance contract, there remain gains from trade between the insurance company and the insuree. Why doesn't the insurance company turn around and offer a full-insurance contract that makes higher expected profits for it and leaves the consumer no worse off?

The answer is immediate. If the insuree anticipates that this is what the insurance company will do, the insuree, whether high-risk or low-, will choose the contract designed for the low-risk consumer, in anticipation of a further offer of a complete insurance contract (necessarily at better terms than the high-risk consumer is initially offered). The insurance company would do a good deal worse than with the original separating contracts.

While immediate, this answer is also a bit too glib on two grounds. First, it assumes that the consumer anticipates the further offering of full insurance if she elects the partial insurance "intended" for a low-risk

consumer. If the consumer doesn't anticipate that this offer will be made (or, as much to the point, if the firm believes that the consumer won't anticipate this), then the company will expect to have higher profits if it offers insurance in this two-stage manner. You should recall at this point the previous discussion about why it is that our formulation of the firm's problem, where the firm offers a single menu of contracts, may not give us the profit-maximizing marketing scheme for the firm to use. In particular, for a consumer who might not see all the options coming to her in a multistage scheme, our formulation would not necessarily find the profit-maximizing marketing scheme.

In addition, (and the main point of this subsection), even if the consumer does foresee all the options that will be made available (as a function of her earlier choices), we have to wonder whether the firm can constrain itself from offering full insurance to a customer who chooses partial insurance from one of our supposedly optimal menu contracts. Ex post, if partial insurance is selected, the firm knows that the consumer is low risk. It knows that it can make more profits by making a further offer. But it is supposed to have the ability to refrain from acting on this information. If it cannot commit not to act on this information, and if the consumer anticipates this, then the consumer anticipates a further offer if she takes the low-risk partial insurance contract. And the firm, anticipating ex ante that it will act in this way (that is, anticipating its inability to refrain later from a further offer), will not wish to try to separate the consumer types. (But see the small type following.)

We don't mean to suggest by this that the analysis given before is wrong. We simply wish to stress that our analysis depends on the firm's ability to make and stick with a particular set of offers. One reason why the firm may have this ability is that it may be involved in this sort of situation repeatedly, so it has a reputation to protect.⁴

Alternatively, we may wish to consider what happens in optimal contract design where the party designing the contract cannot forebear from renegotiating the contract ex post, after the party with private information has taken some action or made some selection. The analysis of this is tricky, because if precise assumptions are made about opportunities to renegotiate contracts, then one can be led to the conclusion that the insurance firm will offer the optimal separating menu at the last possible instant, just before the time at which the consumer's endowment is realized and after all opportunities for renegotiation have passed. This can be complicated in one of two ways: First,

⁴ Or, in the spirit of our discussion in chapter 9, it may offer to consumers "most favored policyholder" contracts, where any offer made to any policyholder must be made available to all others. (This is relegated to a footnote because it gets a bit ahead of the story, to the next subsection where we discuss an insurance firm insuring many consumers simultaneously.)

there may be no "last instant" at all. This, for example, could come about if the time at which the risk is realized (or not) is random; a loss may be incurred at any moment, but if it doesn't happen now, it may happen later. (If it makes sense to you, consider a formulation where the time at which the consumer sustains a loss is random, with positive probability that the time never comes [the loss is never sustained], and conditional on a loss occurring, the time is exponentially distributed.) Second, we could consider a situation in which the loss situation is repeated. The consumer is either high or low risk in each of several periods (i.e., her type doesn't change), but conditional on her type, she stands to suffer a loss in income in each period independently of losses in other periods. For analysis (in a different context) along the second line, see Laffont and Tirole (1988).

Contracts designed for noninteracting parties: Optimal nonlinear prices

In the example, we were careful to speak of the (single) consumer, stressing the idea that this story was about the design of a contract for a single consumer whose type was unknown to the insurance company. Of course, insurance companies deal not with a single individual but with many individuals of varying types. So we can suppose the insurance company is extending insurance to many consumers, each of whom is subject to endowment risk of the sort we indicated. What sort of contracts should the firm offer in this case to maximize its expected profits?

To answer this question we need to ask two more. Can the firm tell which of the consumers is high risk and which is low? And, if it can, is it able to act on that information in the sense that it can offer a different contract or menu of contracts to each individual according to the individual's type? If the answers to both questions are yes, then the solution is just as in chapter 9: Offer a first-best (profit-maximizing) contract for each individual to each individual on a take-it-or-leave-it basis. But if the answer to either question is no, either because (say, on legal grounds) the insurance company is barred from "discriminating" in its offers or because it can't tell the characteristics of the individual, then the sort of analysis performed above becomes appropriate.

An important caveat should be made here. In the case of insurance, it seems reasonable to imagine that the choice of a contract from a menu by one insuree is unaffected by choices made by other insurees.⁵ This is

⁵ Actually, this is not so reasonable if there is risk that the insurance company will default on its obligations. Buying earthquake insurance from a firm that has a large amount of money at risk in the event of an earthquake may be somewhat less desirable than buying the same insurance from a company that does not have a large stake in this particular segment of the business.

crucial to our mode of analysis. As we analyze what choice one consumer will make from a given menu, we don't want to be forced to consider the choices others are making. This isn't to say that the objective function must be constructed separately for each individual consumer; perhaps it can be (if, for example, the firm wishes to maximize the sum of its expected profits), and then the analysis will be identical to that of the first subsection of this section. But even if the firm's objective function has as argument the entire array of consumer choices (which it would, for example, if the firm had a nonlinear cost structure for the provision of insurance), as long as the self-selection constraints can be constructed "individually," the basic techniques used in this section can be applied.

We can imagine other applications, in contrast, where the choices and actions made by one contractee affect how others in a similar position will fare. We move to such applications in the next section.

But before doing so, we note that the style of analysis in this section is precisely what we want for standard problems of optimal nonlinear pricing. Recall the example of chapter 9: A manufacturing monopolist is selling wholesale to retail monopolists. Different retail monopolists face different demand curves, and so, ideally, the manufacturer would like to make different take-it-or-leave-it offers to the retailers. But if the monopolist is constrained by law from doing this, or if the monopolist is unable to do this because she is unable to tell which demand curve faces a particular retailer, then the methods employed here would be used. For each "type" of retailer, where "type" corresponds to the demand curve the retailer faces, the monopolist designs a specific contract. These contracts are constrained so that each type of retailer prefers the contract designed for him to the contracts intended for other types. Subject to those constraints, the manufacturer designs a profit-maximizing menu and lets the retailers self-identify. In case you didn't do problem 4 of chapter 9 before, you get another chance here; it is repeated as problem 1.

18.2. Optimal contracts for interacting parties

Things get more interesting when we try to design optimal contracts for parties whose actions and choices interact. The examples that appear in the literature tend to involve a lot of mathematics. The mathematics isn't (usually) especially deep or complex, but it can get in the way of understanding the basic issues. So we will only go through the analysis of a toy problem that illustrates many of the important conceptual issues, leaving you to read elsewhere about more realistic applications.

A toy problem

Imagine that some party must procure one hundred units of a particular item. We will call this party "the government," and you may think of a government that is seeking to procure one hundred jet fighter planes (although we will work with a formulation that assumes these hundred units are perfectly divisible). Two firms could supply these units. We index the firms by $i = 1, 2$. Each firm has a simple linear cost structure — constant marginal cost c_i and no fixed costs. But, and this is the complicating feature, the value of c_i for each i can be either 1 or 2, each equally likely. The values of c_1 and c_2 are statistically independent. While each firm i knows its own cost, neither the government nor the other firm knows the costs of the firm i . Instead, both the government and the other firm believe that c_i is either 1 or 2 with probability $1/2$ apiece.

We assume throughout that the government is able to propose contracts to the two firms to which they must respond on a take-it-or-leave-it basis. And we assume that they will respond favorably whenever it is (weakly) in their interest to do so.

If the government knew the costs of the two firms, its procurement problem would be easily solved. It would propose to buy from either firm if they have the same unit costs (or split its order between them) and to buy from the lower-cost firm if one firm has unit cost 1 and the other has cost 2. Since the government makes take-it-or-leave-it offers, it need never pay more than the unit costs of the firm, which means that it pays 100 with probability three-quarters and 200 with probability one-quarter, for an expected cost of 125.

But since the government doesn't know the unit costs of the firm, it can't do this. Suppose it tries the following scheme: The firms are asked (voluntarily) to reveal their costs. If both firms name the same cost, the government will split the order between the two, paying the named cost. If one firm names lower costs, the government will buy from the firm naming the lower cost, at that cost. In response to this, if firms are restricted to naming either 1 or 2 as their cost, they will name 2 regardless of what their cost is. When a firm's cost is 2, there is no point to naming a cost of 1; this will result in a loss of either 50 (if the other firm names a cost of 1) or 100 (if the other firm names 2). And when the firm's cost is 1, it makes zero profits if it says its costs are 1 while, if it names 2 and the other firm names 2 (which the other firm is bound to do with probability at least $1/2$), then the first firm makes 50. So both firms will name 2 for sure, and the government's costs will be 200.

Clearly, the government needs to provide firms with an incentive for revealing that their cost is 1 when it is 1. Consider the scheme: If both